

Repurposing abandoned oil and gas wells for geothermal energy extraction: A review of concepts, challenges, and opportunities

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HIGHLIGHTS

- Repurposing abandoned wells for geothermal energy aids sustainable energy transitions.
- Innovative dual-purpose systems for hydrocarbon and geothermal production proposed.
- Closed-loop systems mitigate risks and improve geothermal energy recovery efficiency.
- Economic and environmental benefits of repurposing wells analyzed comprehensively.

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ABSTRACT

Repurposing abandoned oil and gas wells (AOGWs) for geothermal energy (GE) extraction has emerged as a promising strategy at the intersection of energy transition and environmental sustainability. This paper explores the economic, environmental, and technological dimensions of this innovative approach, aiming to provide further insights into its feasibility and potential impact. In terms of operations, a number of challenges were addressed in detail in converting oil and gas wells into GE sources mainly due to abandoned wellbore diameter, heat loss through the well column, low geothermal gradient, thermal facility cost, and unwanted entry of formation fluid into the well. However, accessing drilling data, well production history, and thermophysical and geological information of the reservoir could ease the conversion of AOGWs into GE sources, all while reducing operational costs compared to standard geothermal reservoirs. To prevent GE loss from abandoned wells, the potential applications of different methods such as in-situ combustion, use of horizontal wells, and increasing the depth of abandoned wells through drilling were scrutinized in terms of having favorable conditions to produce high-temperature fluid. Also, the application of simultaneous production of GE and hydrocarbon fluids from abandoned wells was addressed, enabling us the continued production of hydrocarbons alongside geothermal objectives. Results of this study in terms of detailing advantages, challenges, and practical experiences in converting AOGWs into GE sources could serve as a useful guide for decision-makers in the global oil and gas industry to use the subsurface geo-energy in a more sustainable way.

1. Introduction

Nowadays, the most significant sources of geo-energy in the world are oil, gas, and geothermal reservoirs. To boost our oil/gas supplies and increase production, new technologies are being applied in the oilfield [1]. The global increase in energy demand and reliance on fossil fuels has hindered efforts to reduce greenhouse gas emissions [2]. In 2023, approximately 80 % of the world's total energy consumption came from

fossil fuels, demonstrating the world's continued reliance on these non-renewable resources [3]. Moreover, a 30 % surge in global energy demand is anticipated by 2035, attributed to population growth and economic development [4]. Considering the drawbacks of fossil fuels, geothermal energy (GE) can be an environmentally friendly energy source [5]. GE is currently the only source of energy with the potential to provide uninterrupted clean power to the electrical grid without the need for storage and the wasteful consumption of non-essential resources [6]. Geothermal energy, like wind and solar energy, is

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Nomenclature

a	soil diffusivity	R_c	thermal resistance due to conductive heat transfer through the casings of the well
$c_{p,s}$	rock matrix heat capacity at constant pressure (J/kg.K)	R_{i0}	thermal resistance between the outer pipe and inner pipe
$c_{p,f}$	fluid heat capacity at constant pressure (J/kg.K)	T_{out}	outlet temperature (°C)
C_{fluid}	fluid specific heat (J/kg.°C)	T_r, T_s	temperature of rock (°C)
d	equivalent diameter (m)	T_w	heat exchanger/formation interface temperature (K)
d_0, d_1	$11.747, -1.251 \times 10^{-5}$	r_{bnd}	radius of the radial boundary (m)
d_2, d_3	$3.405 \times 10^{-9}, 5.541 \times 10^{-2}$	H_m	bottom coordinate of the m-layer
d_4, d_5, d_6	$2.750 \times 10^{-4}, -7.198 \times 10^{-2}, 1.078 \times 10^{-4}$	u	average section velocity (m/s)
d_e	hydraulic diameter of injection well (m)	$\vec{u}$	velocity field (m/s)
d_{fr}	fracture element aperture	Y_0, Y_1	zero-order/ first order Bessel function of the second kind
$d\dot{Q}/dz$	rate of heat flow over dz. (Wm^{-1})	z	depth (m)
h_{int}	film convective coefficient ($W/m^2.K$)	z_v	vertical distance from earth surface (m)
h_w	convective heat transfer coefficient ($Wm^{-2}K^{-1}$)	ρ	density of the circulating fluid (kg/m^3)
h_o, h_{wall}	convective heat transfer coefficient ($W/m^2.C$)	ρ_{rock}	rock density (kg/m^3)
H	length (m)	ρ_f	fluid density (kg/m^3)
J_0, J_1	zero-order/ first order Bessel function of the first kind	ω	formation heat capacity ($Jm^{-3}K^{-1}$)
k	height of the equivalent coarse grain (m)	τ_D	dimensionless time
k_t	thermal conductivity of materials	Re	Reynolds number
k_{i0}	heat transfer coefficient between the outer pipe and inner pipe	φ	Porosity, dimensionless
K_{eff}	effective thermal conductivity (W/m.K)	μ	dynamical viscosity coefficient (Pa.s)
K_1, K_2	thermal conductivity (K.W/M)	γ	temperature response dimensionless at the wall of the borehole
Nu	Nusselt number (–)	λ_i	heat conductivity of the pipe material ($Wm^{-1}K^{-1}$)
Pr	Prandtl number	$\lambda_s, \lambda_{rock}$	rock matrix thermal conductivity ($Wm^{-1}K^{-1}$)
Q	heat source (W/m^3)	λ_f	thermal conductivity of fluid ($Wm^{-1}K^{-1}$)
Q_e, Q_{out}	external heat exchange through the pipe wall (W/m^2)	λ	friction factor
Q_{in}	mass flow of injection well (kg/s)	∇_T	refers to the gradient operator limited to the tangential plane of the fracture element
q	heat flux per borehole length (W/m)	Δp_f	frictional pressure drop (Pa)
q_{out}	outlet flow rate (m^2/s)	Δp	pressure difference between injection well and production well (MPa)
q_s	heat sink (or source)		
t	temperature (°C)		
t_p	production time (day)		
T_b	borehole wall		
T_o	initial ground temperature		
T_{fi}	fluid temperature in the inner pipe (°C)		
T_{fo}	fluid temperature in the outer pipe (°C)		
T_f	fluid temperature (K)		
T_{in}	inlet temperature (°C)		
T_{ei}	formation temperature at the finite distance from well axis (K)		
T_{out}	outlet temperature (°C)		
T_r, T_s	temperature of rock (°C)		
T_w	heat exchanger/formation interface temperature (K)		
T_{wall}	temperature of the wall of rock (°C)		
r	radial coordinate (m)		
r_2	inner reduce of the production well (m)		
r_c	external radius of the external casing (mm)		
r_d	radial distance from the central axis (m)		
r_0	radius of the internal casing (mm)		
r_{re}, r_i	inner radius of the casing (m)		
r_b, R_b	borehole radius (m)		
r_{inj}	inner reduce of the injection well (m)		
R_{i0}	thermal resistance between the outer pipe and inner pipe		
T_{out}	outlet temperature (°C)		
T_r, T_s	temperature of rock (°C)		
T_w	heat exchanger/formation interface		

Abbreviation

AOGWs	Abandoned Oil and Gas Wells
GE	Geothermal Energy
BHT	Bottom-Hole Temperature
EGS	Enhanced Geothermal Systems
GWs	Geothermal Wells
AWs	Abandoned Wells
OLSs	Open-Loop Systems
AGS	Advanced Geothermal Systems
CLSs	Close-Loop Systems
DPHE	Double-Pipe Heat Exchanger
UTHE	U-Tube Heat Exchanger
BHEs	Borehole Heat Exchangers
DBHEs	Deep Borehole Heat Exchangers
LCOE	Levelized Cost of Electricity
LCOH	Levelized Cost of Electricity
NPV	Net Present Value
CHP	Combined Heat and Power
AW	Abandoned Well
STES	Subsurface Thermal Energy Storage
BOP	Blowout Preventer
DHSV	Downhole Safety Valve
PSAC	Petroleum Services Association of Canada

considered a renewable and clean energy source and can be utilized day and night. Many countries have used this energy source for electricity generation, and it is expected to continue growing, with an estimated production of 800–1300 TWh of electricity and direct heat consumption of 3300–3800 TWh per year by 2050 [7]. Fig. 1 illustrates the installed geothermal capacity and its production trend over the past 43 years. The data indicate that installed geothermal capacity has increased nearly eight-fold, rising from 2110 MW in 1980 to 16,318 MW by 2023. This expansion reflects significant advancements in geothermal technology and a growing recognition of its potential application to enhance energy sustainability. Similarly, annual geothermal production has experienced considerable growth (about 7.4 times), rising from an estimated value of 13,100 GWh in 1980 to 96,552 GWh in 2023. This sustained growth in both capacity and production emphasizes the importance of the geothermal sector in diversifying energy resources and reducing reliance on fossil fuels. The data highlights the critical need for ongoing investment in geothermal technologies, positioning GE as a key player in future energy strategies and emissions reduction efforts [8].

GE is an internationally accessible asset, located at depths spanning from 3 to 10 km underneath the earth's surface [9]. However, the geothermal gradient fluctuates from one area to another. Thermal gradient within the range of 4 to 7.3 ($^{\circ}\text{C}/100\text{ m}$) facilitates reaching bottom-hole temperature (BHT) between 40 and 73 ($^{\circ}\text{C}$) for a well with a depth of 1000 m. This range of rock temperature is fitting for direct implementation of enhanced geothermal system (EGS) at lower temperature, including daily applications such as district heating, greenhouse heating, and aquaculture [10]. Typically, four distinct categories encompass GE sources: hydrothermal resources, hot dry rock, geopressured (influenced by pressure from earth's crust), and magma resources. A geothermal field is commonly denoted as a hydrothermal or geothermal reservoir, where the unceasing natural flow of heat and fluid occurs. These fields are predominantly characterized by water or steam, and the most viable energy derived from them can be employed for either electricity generation or direct applications. Following the extraction of thermal energy, the working fluid is reinjected into the ground through reinjection wells to cyclically return to the reservoir [11].

In spite of the remarkable capacity of geothermal energy, advancements in its application and acceptance have proceeded at a measured pace, primarily centered in particular geographical zones characterized by substantial volcanic and thermal activities. The primary factor contributing to this sluggish advancement is the demanding, high-stakes, and expensive process of drilling geothermal wells (GWs). A potential solution to overcome this challenge is the use of deep

underground mines or abandoned oil and gas wells (AOGWs). The presence of resources from oil/gas can contribute to the advancement of geothermal technologies by virtue of the shared characteristics between the two industries in domains like professional expertise, governmental organization, political alliance, underground information, and material infrastructure [12]. However, the reliability of geothermal resource potential estimates in oil and gas fields remains uncertain [13]. To address this, Table 1 provides a comprehensive summary of the previous studies that explore the utilization of AOGWs for GE development, offering valuable insights into the current state of research and practical applications.

The number of abandoned wells (AWs) is increasing toward the end of the oil field development period [22]. Globally, around 20 to 30 million oil/gas wells have been abandoned [23]. Gas wells that have been left unused possess substantial potential for transformation into GWs. Generally, these wells come with comprehensive reports and data spanning their entire production lifespan, enabling a comprehensive evaluation of well performance, risk mitigation, and cost-effectiveness. This leads to a reduction in resource wastage and has several economic benefits. Furthermore, existing tools in the oil/gas industry can be used to evaluate the GE production potential from mature oil/gas fields, utilizing the heat present in the produced fluid. Millions of globally deserted oil/gas wells have been left inactive, with a significant number releasing pollutants into the atmosphere and aquifer [24–26]. As the oil/gas field enters its final stages of development and production, the challenges of operation are increasing. If AOGWs can be repurposed for harnessing geothermal energy, safety risks or fluid leakage problems resulting from improper well plugging during the abandonment phase can be addressed, thus preventing additional costs [13,27]. Accordingly, in some places, hydrocarbon fields come with conveniently accessible geothermal resources of low to medium enthalpy that can be feasibly reached and the geothermal gradient and subsurface heat flow rates are relatively high [13]. Also, as to numerous reports, many mature oil and gas wells worldwide produce water with temperature ranging from 65 to 150 ($^{\circ}\text{C}$) [28]. Conventional hydrocarbon reservoirs typically house a deep, pressurized aquifer with high temperature underneath hydrocarbon layers, guiding the flow of oil and gas toward production wells. As production continues, the reservoir experiences depletion, leading to the extraction of the significant amount of water alongside hydrocarbons. Beyond subsurface aquifers, unconventional reservoirs like shale or coalbeds, coupled with production techniques such as water flooding, steam injection, hydraulic fracturing, and other methods, can impact both the quantity and quality of the produced water [29]. The water produced from hydrocarbon reservoirs, due to its challenging

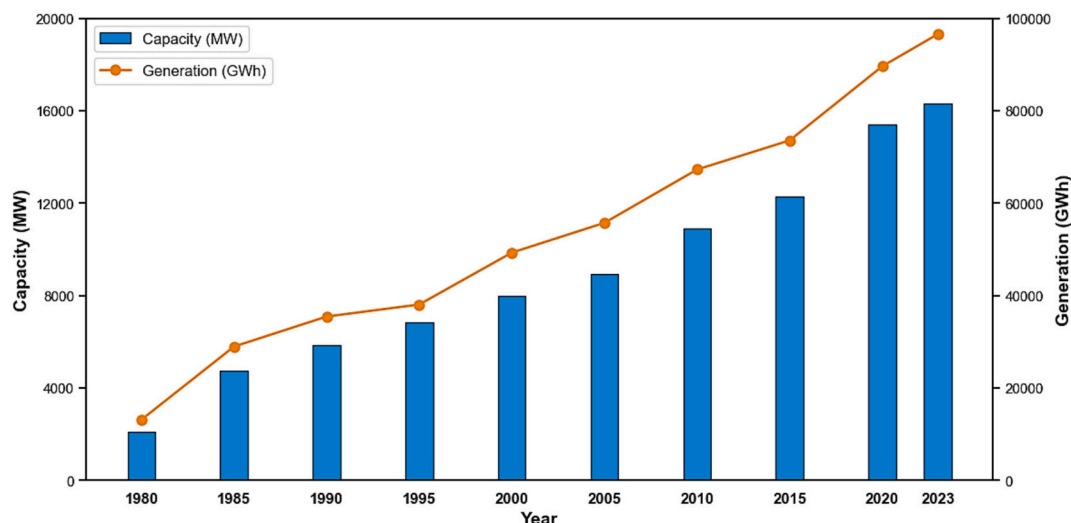


Fig. 1. Global geothermal capacity and electricity generation from 1980 to 2023 (modify after) [8].

Table 1

A thorough overview of previous studies on repurposing oil/gas wells for geothermal development.

Technological Advancement Description	Field Name	Main Objective	Ref
Utilizing AOGWs for GE extraction through the development of a thermo-economic model using the Organic Rankine Cycle (ORC) and analyzing various working fluids enables the economic and efficient use of low-temperature geothermal resources, helping to reduce cost and increase efficiency in GE production.	Shushufindi oil field (Ecuador)	Explore the feasibility, efficiency, and economic viability of repurposing these wells for GE production, offering insights into the potential benefits and challenges of such a transition.	[14]
Reusing oil and gas wells for gas storage and heat and power production through combined heat and power (CHP) plants-Producing biogas from organic waste as a renewable energy source- replacing completions and pumping systems, using Combined Heat and Power (CHP) Systems.	Ragusa oil field (Italy)	Improving oil and gas infrastructure with renewable and low-carbon resources and green biogas production	[15]
Utilized fractured carbonate reservoirs and advanced thermal modeling to optimize GE extraction.	Huabei Oil Field, China	Conversion of AOWs to GE production.	[16]
Utilizing hot water extracted from the geothermal reservoir to power surface chillers for the space-conditioning system. The hot water is pumped to the surface, stored in an insulated tank, and then circulated through pumps to the HVAC ¹ system to cool the air, which is then delivered to the desired space.	RELLIS lease area (Shale field)	Technical/economic challenge of converting abandoned shale wells to GWs	[17]
Installing thermoelectric generators (TEG) on the outer side of the production string. These TEGs convert thermal energy from hydrocarbon reservoir fluids into electric power, enhancing thermal recovery efficiency and generating electricity directly downhole.	Simulated model	Downhole geothermal power generation in oil and gas wells	[18]
Staged fracturing for horizontal wells is used to stimulate the reservoir, allowing oilfield-produced water to circulate through the fractures and generate electricity from enhanced geothermal systems (EGS).	Daqing Oilfield (China)	Feasibility of generating electricity from EGS by oilfield	[19]

Table 1 (continued)

Technological Advancement Description	Field Name	Main Objective	Ref
Using a multigeneration system, high-pressure water is injected into the well, and as it descends, it transforms into steam. This high-pressure steam is then extracted from the well.	Pirkoh gas field (Pakistan)	Using AOGWs for multigeneration energy production and converting geothermal units into self-sufficient and environmentally friendly units.	[20]
Use of heat exchanger with coaxial thermal insulation in deep oil well	Simulated model (Ahvaz oil well)	Explore the possibility of utilizing repurposed oil wells.	[21]

¹ Heating, Ventilation, and Air Conditioning

management, significantly escalates operational costs and ultimately leads to well abandonment or even field abandonment. In the process of hydrocarbon production, there are numerous AOWs extracting substantial amounts of water. Furthermore, dry exploration wells can also be abandoned, in which case the opportunity arises to repurpose such wells for geothermal conversion [30]. Generating electricity from such AOWs using water as the circulating fluid in the well provides the potential for greater utilization. Typically, water is selected as the circulating fluid due to its high heat capacity, heat transfer capabilities, good thermal conductivity, availability, and also environmental issues. In tight gas reservoirs, supercritical carbon dioxide performs better performance and is recommended instead of water [31]. Also, steam can be used as another working fluid in redesigned hydrocarbon wells to effectively harness the heat from GWs in a multi-purpose system [20]. Accordingly, the upstream oil and gas industry needs to examine how it can engage with these new energy resources and preserve its share in the energy mix by combining geothermal and hydrocarbon resources [6]. It is important for oil and gas companies to begin diversifying their activities and expanding their portfolio of sustainable energy technologies in order to remain key players in the energy sector [32].

1.1. Objective and scope of the review

While previous studies have explored different aspects of GE extraction from oil and gas reservoirs [33,10,13,30,34–43], there still remains a significant gap in addressing the potential for repurposing AOGWs into GWs. Table 2 provides an overview of the previous studies on the geothermal energy recovery from abandoned wells, highlighting the main contributions, limitations, and how this review addresses these gaps. This review aims to bridge that gap by providing a detailed evaluation of the technical, economic, and environmental aspects of converting AOGWs into GWs. It also emphasizes the challenges and opportunities associated with utilizing these wells for dual-purpose energy production, enabling the simultaneous extraction of hydrocarbons and geothermal energy. The primary issues addressed in this review paper include:

- The lack of integrated studies that comprehensively examine the conversion of AOGWs into GWs, considering diverse well profiles, geological settings, temperature conditions, and their implications for energy recovery efficiency.
- The absence of systematic evaluations of the key technical, economic, and environmental criteria required to determine the feasibility and potential of AOGWs for GE production, including factors such as geothermal gradient, well depth, stable temperature outputs, and retrofitting requirements.
- The challenges related to dual-purpose energy production, particularly in maintaining wellbore integrity, optimizing fluid flow, and enhancing heat recovery, which are crucial for efficient geothermal energy extraction. This approach maximizes the utility of AOGWs

Table 2

Overview of previous studies on geothermal energy recovery from abandoned wells, highlighting contributions and limitations.

Ref	Main Focus	Aspects Covered	Shortcomings/Gaps	How Current Review Addresses Gaps
[33]	Review of geothermal energy as a co-product from oil and gas operations including extraction methods and case studies.	Technical (extraction methods, co-production); Economic (feasibility in oilfields); Environmental (reduced emissions via co-use).	Lacks comprehensive integration of diverse well profiles/geological settings; minimal focus on dual-purpose (simultaneous hydrocarbon + geothermal) production challenges; insufficient coverage of regulatory/decommissioning concerns; no future directions on AI/ML or advanced materials.	Provides integrated studies across diverse profiles/settings; systematic evaluation of technical/economic/environmental criteria; detailed challenges in dual-purpose production (e.g., wellbore integrity, fluid flow); covers environmental/economic implications (costs, policies, regulations); includes future directions (AI/ML, materials, simulations).
[10]	Investigation into suitability of abandoned wells in California for Enhanced Geothermal Systems (EGS) and low-temperature applications.	Technical (EGS suitability, heat extraction); Economic (cost savings from reuse); Environmental (reduced drilling impact).	Region-specific (California); limited systematic evaluation of global criteria like geothermal gradient/well depth; no emphasis on dual-purpose production or retrofitting challenges; overlooks broader economic implications like market demand/policies.	Bridges gaps with global, comprehensive approach; systematic criteria evaluation (gradient, depth, outputs); addresses dual-purpose challenges (integrity, flow, costs); delves into environmental/economic aspects (capital/operational costs, incentives, frameworks).
[13]	Comprehensive review of low-temperature geothermal extraction from hydrocarbon fields including methods and challenges.	Technical (extraction techniques); Economic (viability in fields); Environmental (low-carbon integration).	Primarily low-temperature focus; lacks integration of diverse geological/temperature conditions; minimal on dual-purpose systems; insufficient on decommissioning/regulatory concerns; limited future research directions.	Expands to all temperature ranges/diverse conditions; integrates dual-purpose opportunities/challenges; provides systematic evaluations; covers full environmental/economic implications; outlines future directions (AI/ML, materials, hybrid systems).
[30]	Techno-economic and policy review of geothermal extraction from abandoned wells including modeling and barriers.	Technical (modeling, systems); Economic (costs, feasibility); Policy/Environmental (barriers, sustainability).	Strong on policy but limited integration of well profiles/settings; overlooks dual-purpose production complexities; insufficient on heat recovery enhancements or corrosion/scaling; no emphasis on future tech like AI/simulations.	Offers comprehensive integration of profiles/settings; addresses dual-purpose challenges (technical complexity, costs); explores practical considerations (heat-exchangers, fluids, corrosion); provides actionable future recommendations (AI/ML, simulations, materials).
[34]	Review of technologies and numerical simulations for geothermal energy extraction from abandoned wells.	Technical (numerical modeling, heat exchanger systems); Economic (simulation-based cost optimization).	Heavy focus on numerical simulations; limited integration of diverse geological settings; minimal discussion on dual-purpose production; lacks comprehensive environmental or regulatory analysis; limited future research directions beyond simulations.	Integrates diverse geological settings; addresses dual-purpose production challenges; provides comprehensive environmental/economic/regulatory evaluations; proposes future research directions including AI/ML, advanced materials, and hybrid systems.
[35]	Review of utilizing existing hydrocarbon wells for geothermal systems focusing on technical and economic feasibility.	Technical (retrofitting, EGS, heat extraction); Economic (cost-benefit analysis); Environmental (sustainability benefits).	Limited to existing wells, less focus on abandoned wells; minimal emphasis on dual-purpose production; lacks detailed regulatory or decommissioning analysis; limited future tech directions (e.g., AI, materials).	Specifically targets abandoned wells; emphasizes dual-purpose production; includes detailed regulatory/decommissioning considerations; proposes advanced future directions (AI/ML, materials, smart systems).
[36]	Review of geothermal extraction and utilization in oilfields focusing on technologies and applications.	Technical (extraction/utilization tech); Economic (oilfield integration); Environmental (reduced waste).	Oilfield-centric; limited on abandoned well repurposing; no deep dive into dual-purpose challenges; insufficient on broader economic/regulatory frameworks; lacks future directions on advanced tech.	Focuses specifically on AOGW repurposing; emphasizes dual-purpose opportunities; provides integrated technical/economic/environmental framework; includes future research (AI/ML, materials, CCS integration).
[37]	Insights into geothermal utilization of abandoned wells including simulations and optimizations.	Technical (simulations, heat extraction); Economic (cost reductions).	Simulation-heavy; lacks comprehensive environmental assessment; minimal on dual-purpose or regulatory challenges; no emphasis on diverse geological settings or future innovations.	Builds on simulations with comprehensive approach; addresses environmental implications; covers dual-purpose challenges; integrates diverse settings and future directions (AI, materials, smart systems).
[38]	Current status and future prospects of repurposing abandoned wells for geothermal energy including technical and policy aspects.	Technical (retrofitting, well integrity); Economic (feasibility); Environmental (sustainability, reduced emissions); Policy (regulatory barriers).	Limited integration of diverse geological/temperature conditions; minimal focus on dual-purpose production challenges; insufficient coverage of advanced technologies like AI/ML or materials; less emphasis on economic implications like market demand.	Integrates diverse geological/temperature settings; emphasizes dual-purpose production challenges; proposes advanced technologies (AI/ML, materials); covers comprehensive economic implications (market demand, incentives, decommissioning).
[39]	Fundamentals applications and research on thermal potential of abandoned wells.	Technical (fundamentals, applications); Economic/Environmental (research overviews).	Broad but lacks detailed systematic criteria for feasibility; minimal focus on dual-purpose production; insufficient on economic implications like market demand or decommissioning costs.	Provides detailed systematic evaluations; emphasizes dual-purpose energy production; delves into full economic/environmental/regulatory challenges; synthesizes gaps with actionable recommendations.
[40]	Heat transfer characteristics and economic analysis of closed-loop coaxial heat exchangers in abandoned wells.	Technical (heat transfer, coaxial systems); Economic (cost analysis); Environmental (sustainable reuse).	Focus on specific system (coaxial heat exchanger); limited to closed-loop systems; minimal dual-purpose production discussion; lacks broader regulatory or future tech focus.	Expands to both open/closed-loop systems; integrates dual-purpose production; includes regulatory and broader economic evaluations; proposes future directions (AI, hybrid systems, materials).

(continued on next page)

Table 2 (continued)

Ref	Main Focus	Aspects Covered	Shortcomings/Gaps	How Current Review Addresses Gaps
[41]	Assessment of geothermal potential in Alberta using analytical models and simulations.	Technical (conjugate model, Monte Carlo simulations, screening criteria); Economic (turning assets profitable); Environmental (reducing carbon footprint).	Region-specific (Alberta); limited to modeling without broad regulatory or dual-purpose emphasis.	Addresses diverse geological settings; systematic criteria evaluation; covers dual-purpose and environmental/economic implications; adds future research on simulations and AI.
[42]	Economic feasibility environmental impact and strategies for repurposing depleted wells with case studies.	Technical (drilling tech, EGS, materials); Economic (high costs, LCOE, job creation); Environmental (low emissions, water management).	High initial costs; location-dependency; EGS in developmental stages; water resource stress in arid areas.	Bridges cost and location gaps with comprehensive economic evaluations; integrates dual-purpose to enhance feasibility; provides framework for sustainability, including advanced materials and AI for predictive modeling. Addresses limitations with detailed heat recovery and dual-purpose strategies;
[43]	Feasibility in Oman conventional and EGS applications.	Technical (energy yield, EGS for heat extraction); Economic (financial viability); Environmental (decarbonization).	Limited power output from conventional applications due to well constraints; need for alternatives like EGS.	systematic technical criteria; covers broader economic and regulatory challenges; outlines future optimizations via simulations and hybrid systems.

but provides unique challenges such as technical complexity, engineering issues, and high retrofitting costs.

- The insufficient coverage of environmental and economic implications, including the capital and operational cost of repurposing wells, market demand, policy and fiscal incentives, regulatory frameworks, and decommissioning concerns.
- Future research directions include integrating artificial intelligence (AI) and machine learning (ML) for predictive modeling, developing advanced materials to improve wellbore integrity, and simulating energy production under different system configurations to enhance performance and efficiency.

Thus, the goal of this review mainly lies in its comprehensive approach to repurposing AOGWs for geothermal energy production. It also addresses technical aspects such as geothermal wellbore integrity, heat recovery potential, and other relevant technical evaluations, while also exploring practical considerations such as heat-exchanger technologies, working fluids, corrosion, and scaling. Moreover, this paper delves into the environmental, economic, and regulatory challenges of repurposing AOGWs, providing an integrated framework for assessing

the feasibility and sustainability of this innovative energy source. By synthesizing existing knowledge and identifying key gaps, this paper provides valuable insights and actionable recommendations for future research and development. It aims to optimize the use of AOGWs in geothermal energy generation, while also exploring their potential for dual-purpose energy production. This forward-looking perspective ensures that the paper not only consolidates current knowledge but also serves as a foundation for advancing the field through innovative research and practical applications.

2. Systems employed for generating geothermal energy from abandoned wells

As the geothermal potential varies across different regions of the earth, the selection of optimal well and system configurations is contingent upon the specific geothermal characteristics of each site. A variety of technologies has been innovated to harness GE from wells that have been abandoned, as shown in Fig. 2. In enhanced geothermal systems, GE is extracted using open-loop systems (OLSs), whereas advanced geothermal systems (AGS) employ closed-loop systems (CLSs)

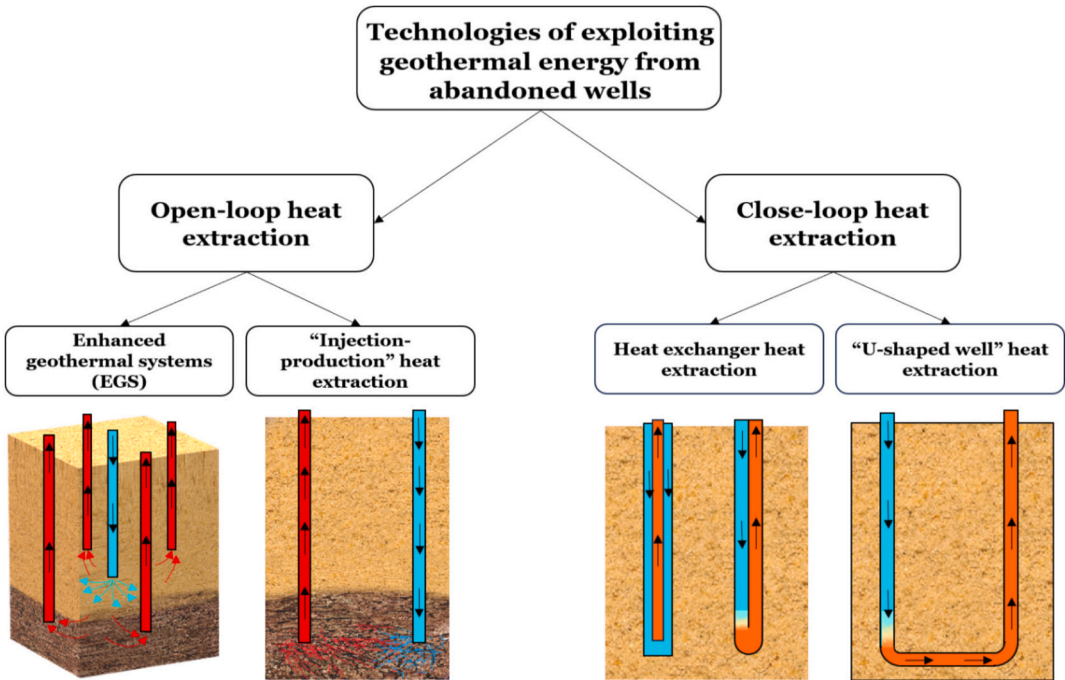


Fig. 2. Classification of different technologies for exploiting GE from repurposing wells.

[6,44]. Closed-loop geothermal systems offer a suitable alternative to OLSs, primarily due to their ability to mitigate issues such as loss of working fluid, surface settlement, formation compaction, and seismic activity. However, it is important to note that CLSs generally exhibit lower heat recovery efficiency compared to OLSs. To enhance heat recovery from closed-loop systems, the incorporation of hydraulic fracturing techniques has been proposed [45]. This approach aims to improve the heat production rate in closed-loop geothermal systems, as depicted in Fig. 3.

The available reports have shown a growing interest in CLSs in comparison to OLSs. Open-loop geothermal systems utilize multiple wells to circulate a heat-carrying fluid through a geothermal reservoir, necessitating substantial upfront investment for well drilling and ongoing brine extraction. This approach presents several challenges, including pipe corrosion, land subsidence, gas emissions, and potential water pollution. For instance, the precipitation of halite can occur due to cooling in brine, particularly when production approaches near-saturation conditions. In contrast, closed-loop geothermal systems operate using a single well equipped with downhole heat exchangers to extract heat energy from rock formations. In this configuration, a heat-carrying fluid circulates within the wellbore heat exchanger, enabling heat extraction without direct contact between the fluid and the surrounding rock. This design effectively addresses many of the challenges associated with OLSs [46,47]. CLSs function as heat exchangers, allowing for heat extraction while significantly reducing gas emissions into the atmosphere. Additionally, this system minimizes environmental impacts by eliminating the need for geothermal fluid extraction from the reservoir, which also helps mitigate corrosion issues since the circulating fluid does not interact directly with the reservoir. However, a notable drawback of CLSs is their reduced efficiency and lower electrical power generation when compared to conventional geothermal power plants, particularly in terms of heat recovery in heat exchangers.

In a CLS, heat exchangers can be employed, and two frequently used models including the double-pipe heat exchanger (DPHE) (Fig. 4 (A)) and the U-tube heat exchanger (UTHE) (Fig. 4 (B)). The U-tube heat exchanger is commonly employed in shallow formations, particularly for heating purposes. In contrast, the double-pipe heat exchanger, with a larger heat transfer surface compared to the U-tube type, theoretically provides superior performance in heat extraction. For AOWs, a double-pipe heat exchanger is a better choice than a U-shaped one. Another option is the spiral/tubular heat exchanger, but its practical implementation in the limited space of AWs can be challenging.

Fig. 5, right section, illustrates different heat exchanger examples, and the left section shows the closed-loop well structure where fluid is injected into the annular space and directed upward through the production pipe in a closed-loop manner [48,49]. Among borehole heat exchangers (BHEs), the coaxial BHE is expected to perform well at medium depths, while the U-tube BHE type is more suitable for use in homogeneous reservoirs [36,50,51]. Old AWs can be revitalized for heating purposes based on shallow or much deeper borehole heat exchanger systems [50]. Fig. 6 illustrates BHE types based on their thermal resistance. Different studies have conducted comparative analyses of the heat transfer properties of BHEs due to the variation in their thermal performance. Specifically, they have focused on creating models to estimate the thermal resistance (R) of BHEs, which comprises the resistance between pipes and surrounding rocks, as well as the resistance of each pipe and the surrounding rocks. For the shallow depth of AOGWs, converting these wells into GWs in the form of a closed-loop heat exchanger is usually used [52–54]. The coaxial borehole heat exchanger, which consists of two concentric tubes, offers lower drilling costs compared to the U-tube design [52]. Doublet deep borehole heat exchangers (DBHEs) demonstrate superior performance in extracting GE from AWs when compared to single-well DBHEs, such as U-tube heat exchangers. Additionally, doublet DBHEs achieve higher operational

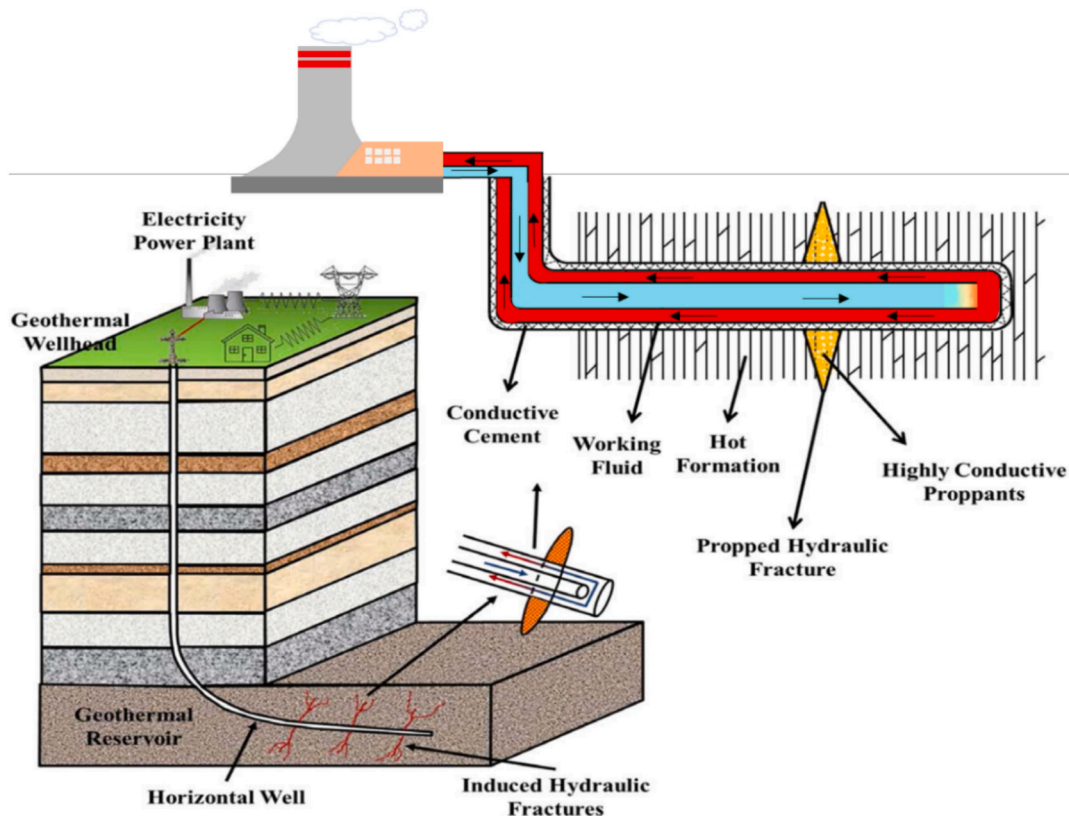


Fig. 3. A closed-loop horizontal geothermal system with hydraulic fractures filled with high conductive material to enhance the efficiency of heat extraction from the reservoir (modify after) [46].

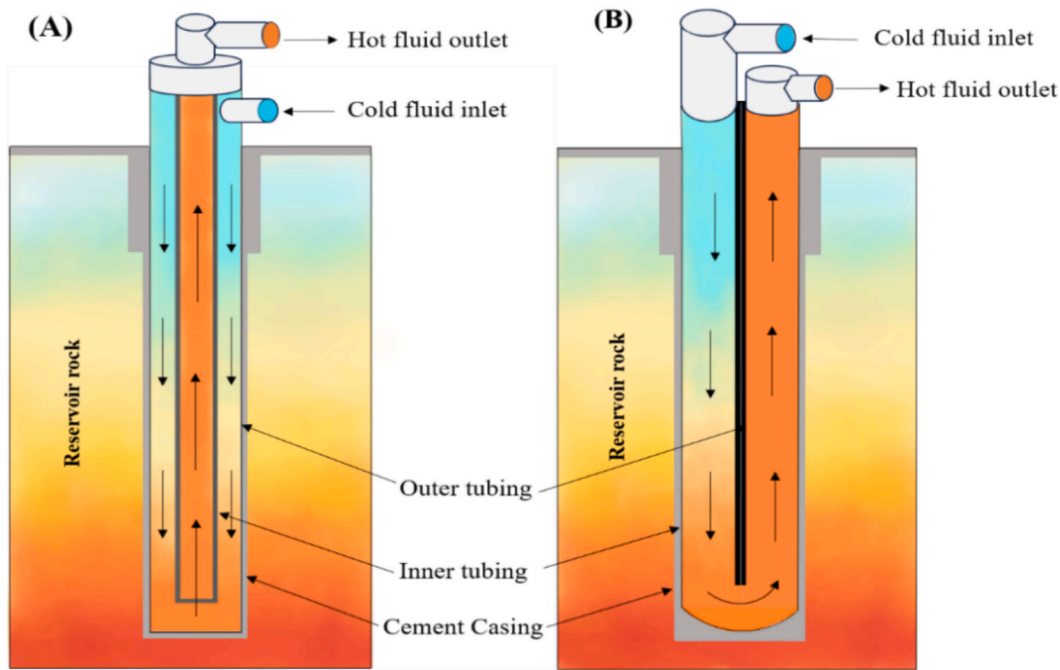


Fig. 4. Configuration of the heat exchanger used for EGS. (a) Coaxial pipe or double pipe type and (b) U-tube type.

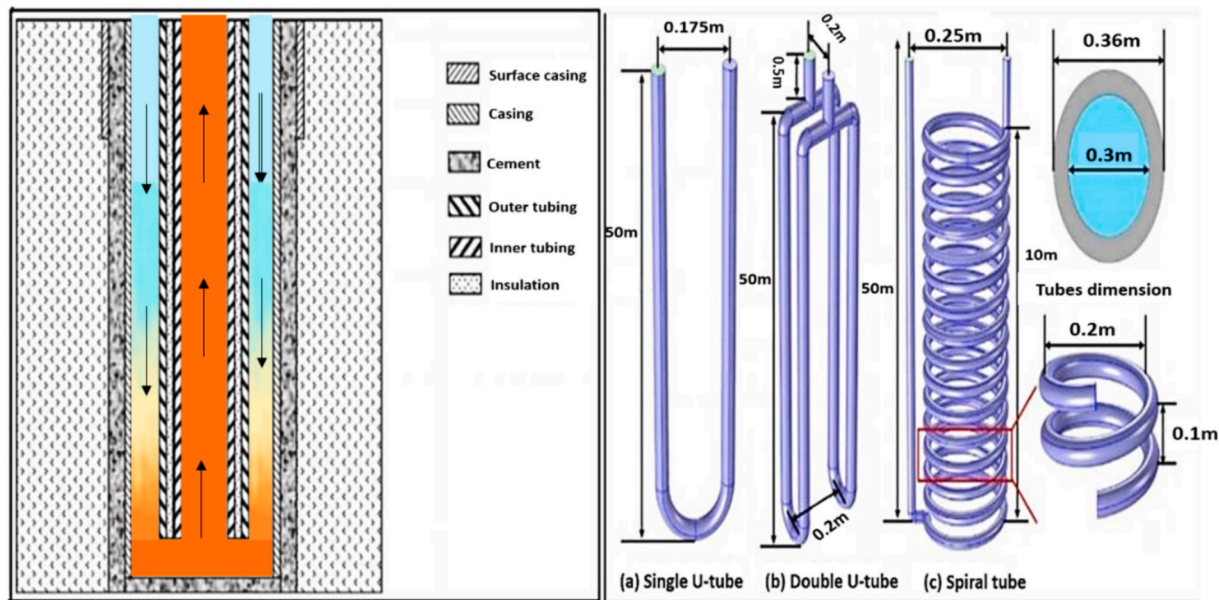


Fig. 5. Right side; Types of heat exchangers. Left side; Closed loop well structure (modify after) [47].

temperature and greater output power [53]. In U-tube heat exchanger systems, a high injection rate results in a reduction of the produced fluid's temperature. At a constant inlet velocity, increasing the diameter of the U-tube pipe leads to a decrease in the outlet water temperature while enhancing the overall heat extraction rate [54]. As an application example, in the Croatian section of the Pannonian Basin, within the Pannonian sub-basin, there is an abandoned deep well named "Pčelić," which is equipped with a coaxial heat exchanger that has been utilized at the shallow depth. The design of the well, including the fluid flow direction and heat transfer mechanisms, is illustrated in Fig. 7. The proposed fluid flow involves injection into the inner pipe and extraction through the outer pipe. In this GE setup, the fluid at the lowest temperature is heated by the surrounding rock formations, reaching its

maximum temperature and heat transfer efficiency at the bottom of the well [55].

Table 3 provides an overview of deep BHEs located in Germany and Switzerland. These heat exchangers were installed in AOWs after their production phase had concluded. The systems utilize a concentric pipe configuration, with one pipe facilitating the downward flow of fluid, while the other supports the upward flow of heated fluid toward the surface. Once the fluid reaches the surface, it is directed through heat pumps to the consumption areas. The effectiveness of this deep BHE configuration has been validated and implemented for direct heating applications [10]. The scenario of electricity production from GE via borehole heat exchangers, without the extraction of geothermal fluids, has been recognized as an economically optimal approach, considering

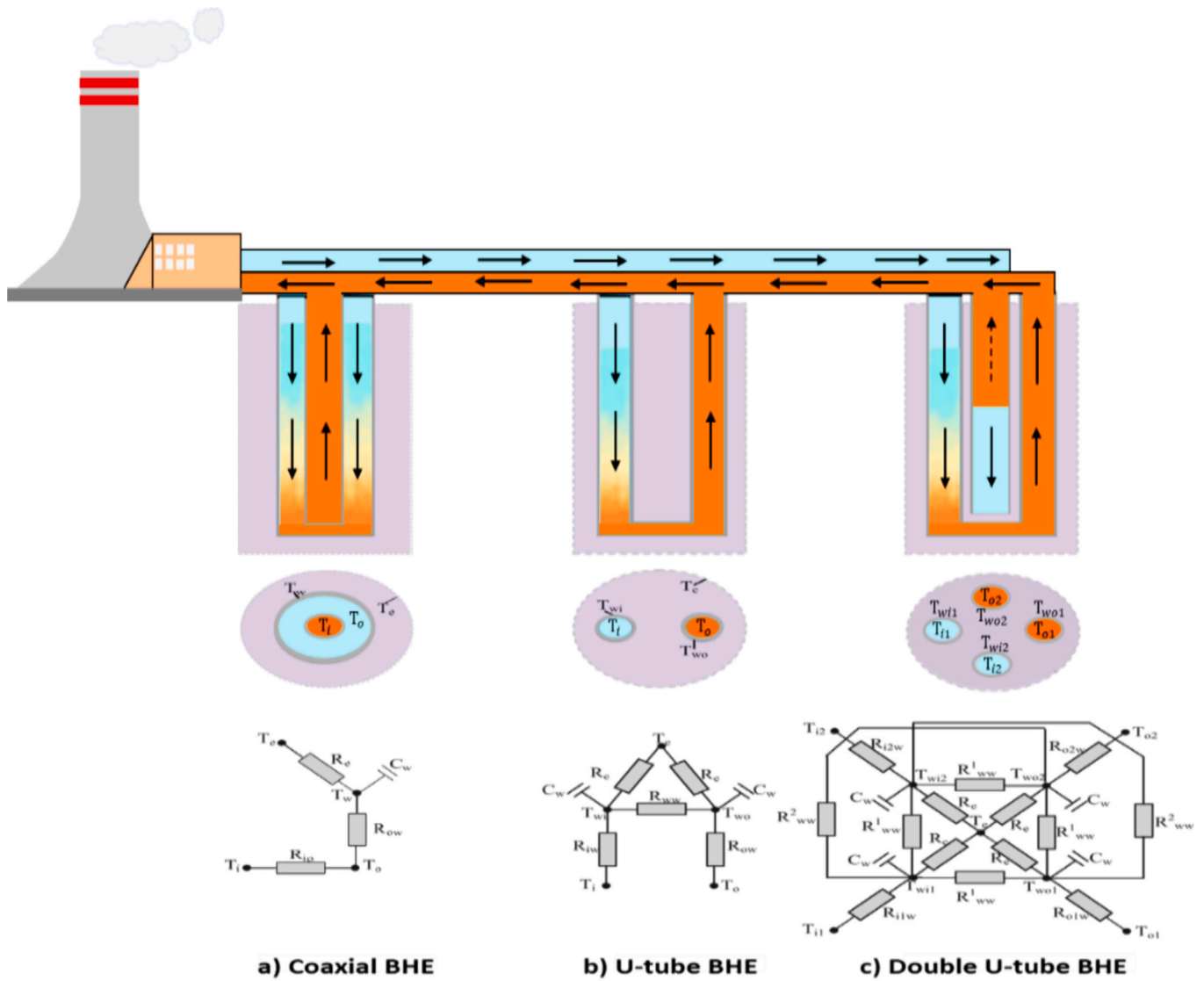


Fig. 6. A modeling approach for different types of borehole heat exchangers (BHEs) based on their thermal resistance (modify after) [37].

parameters such as the levelized cost of electricity (LCOE), levelized cost of heat (LCOH), and net present value (NPV). Furthermore, the combined heat and power (CHP) scenario represents an advanced method, integrating heat and power production through parallel well configurations. This scenario involves two operational modes: the first mode incorporates production and injection wells, while the second utilizes BHEs. The CHP scenario is recognized as one of the most effective strategies for transitioning from oil production to water production, especially in wells with a significant water cut. This approach not only optimizes production rates but also contributes to reducing CO₂ emissions [56].

3. Key parameters and effective approaches for repurposing abandoned wells into geothermal wells

The parameters that influence total electricity production when utilizing GE from AOGWs include fluid flow rate, well length, internal and annular pipe diameters, geothermal temperature, type of working fluid, and overall heat transfer coefficients [27]. However, within the reservoir, geological factors have a significant influence on the status and quality of key geothermal parameters such as temperature, the occurrence of subsurface reservoirs, the physical and chemical properties of water, rock porosity, permeability, and more [57]. The

temperature of the produced geothermal fluid at the surface and the resulting output power are significantly affected by two key factors: the temperature of the injection water and the geothermal gradient. Fluid flow rate also plays a critical role in heat exchange and in determining the overall output power [26,56]. Bu et al. (2012) [58] conducted a feasibility assessment on harnessing GE from AOGWs. Their study revealed that a well must have a minimum depth of 4.2 km and a thermal gradient of 45 °C/km to be viable for both direct heat applications and electricity generation. Therefore, utilizing GE becomes challenging when the well's depth and thermal gradient fall below these thresholds. Temperature differences of over 30 °C can exist between wells that are 2 km and 5 km deep, significantly affecting the feasibility of using AWs and geothermal units. Furthermore, in AOGWs converted to GWs, a minimum distance of 40 m between two GWs should be maintained to avoid interference. The use of nanofluids, particularly Al₂O₃/water, Cu/water, and CuO/water, with particle volume concentrations ranging from 1 % to 4 % in a coaxial wellbore heat exchanger, has been shown to enhance the outlet temperature of the fluid from the well. For example, incorporating 4 % Cu nanoparticles into water results in a 25 % increase in the fluid's exit temperature, demonstrating superior thermal performance compared to other nanofluids [23].

A significant challenge in GE extraction is the low heat transfer rate, with increased rates often leading to higher pressure drops. High mass

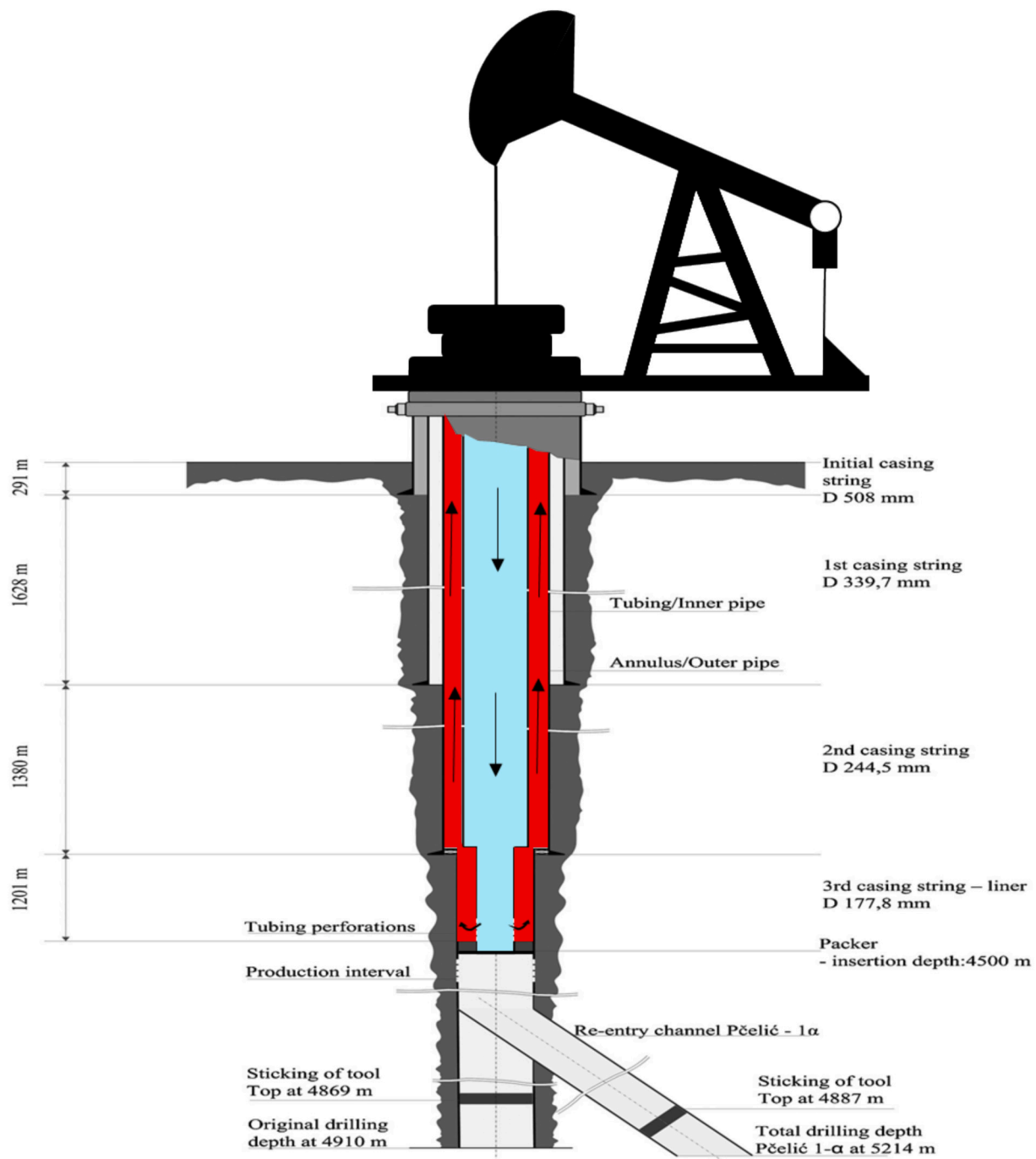


Fig. 7. The schematic of “Pčelić-1” geothermal well and its design, including the installation of a packer to determine the fluid flow direction (modify after) [55].

Table 3
Existing deep borehole heat exchanger sites [10].

Location, Country	Peaking method	EWT ¹ °C	Depth (m)	Flow rate (l/s)
Aachen, Germany	Heat pump	25–55	2500	2.77
Prenzlau, Germany	Heat pump and Gas/oil boiler	–	2786	6
Weissbad, Switzerland	Heat pump	15	1200	2.9
Weggis, Switzerland	Heat pump	40	2300	0.8–1.75

¹ Entering water temperature (EWT) is outlet temperature from repurposed well heat exchanger.

flow rates of fluid can deplete the bottom-hole heat source, disrupting the GE extraction process. To ensure stable production, it is essential to consider the optimum injection temperature and the minimum injection pressure required for system operation. Due to lower outlet temperature, the practical application of GE extracted from repurposed hydrocarbon wells is less common compared to conventional GWs. For direct heat applications, the location of wells needs to be close to energy consumers to minimize heat losses during distribution. Transporting hot fluids from wells to the designated area of use poses a challenge, particularly due to heat losses in pipelines during distribution [10,38,59]. In addition to exploiting GE from the configuration of AWs, when the total number of wells and the overall flow rate remain constant, having fewer production wells in the system results in a higher average output temperature

[60].

Xiao et al. (2022) [61] demonstrated that the use of liquefied natural gas (LNG) as a cold source in closed-loop wells can store more thermal energy, improve energy extraction efficiency, and be utilized for electricity generation. In closed-loop wells, thermal insulation can be used to prevent production temperature loss [62]. Cheng et al. (2014) [63] investigated the effect of insulation on heat transfer between production and injection wells, finding that the fluid output temperature from the production well increases as insulation thickness grows. For example, the fluid output temperature with 0.3 m of polystyrene insulation was 6 °C lower than that with complete insulation. Despite seemingly straightforward technology, insulating the subsea pipeline network in offshore conditions poses a challenging and expensive endeavor. To counteract heat loss at the seabed level, especially when the well's output temperature is suitable, an effective solution involves installing a geothermal power plant near the production site, typically on an abandoned offshore oil and gas platform. Subsequently, this allows for the transfer of GE, either in the form of electricity or more efficient energy storage to prevent heat loss in these exchangers [64]. In Table 4, the heat transfer equations and supplementary relations related to the modeling of GWs using AOWs are presented. Given that many of these equations are based on numerical simulations and theoretical models, this table provides an overview of analytical tools for the design and optimization of GWs.

Table 4
Heat transfer equations and classic cases of GW modeling using AOGWs.

Classification	Heat Transfer Equation	Observation	Ref.
Close-loop (Coaxial heat exchanger)	$\frac{1}{a} \frac{\partial t}{\partial r} = \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial t}{\partial r} \right) + \frac{\partial^2 t}{\partial z^2}$ $\tau = 0, H_{m-1} \leq z \leq H_m, r_b \leq r \leq r_{bnd}$	Heat transfer in a numerical model of a heat exchanger with varying outer pipe diameters, which uses the finite difference method for complex calculations. In this model, the thermal properties of the ground, pipe, water, and backfilling material are dependent on their specific temperature. The frictional pressure drop caused by the viscous force during the fluid flow has been calculated.	[40]
Close-loop (Coaxial heat exchanger)	$\frac{dQ_{d0}}{dz} = 2\pi r_{d0} k_{d0} (T_{fi} - T_{f0}) = (T_{fi} - T_{f0}) / R_{d0}$	The model developed in the oil wells of Italy assumes heat transfer and constant heat flux in the well components, where the heat extraction from the formation equals the heat flux from the outer surface of the well. $\frac{dQ_{d0}}{dz}$ is determined by considering the temperature difference between the outer-pipe and inner-pipe fluids.	[65]
Close-loop (Double-pipe heat exchanger)	$\frac{d\dot{Q}}{dz} = \frac{2\pi\lambda_e(T_{ei} - T_w)}{f(t)}$ $f(t) = \frac{16\omega^2}{\pi^2} \int_0^\infty \frac{1 - \exp(-\tau_D u^2)}{u^3 \Delta(u, \omega)} du$ $\Delta(u, \omega) = [uY_0(u) - \omega Y_1(u)]^2 + [uJ_0(u) - \omega J_1(u)]^2$	Numerical simulation of a 'dry hole' oil well model based on transient formation heat transfer with insulation applied to the outer wall of the inner tubing, assuming that the formation temperature varies linearly with distance from the wellbore in the vertical direction.	[66]
Open-loop	$(\rho_f c_{pf})_{eff} \frac{\partial T}{\partial t} + \rho_f c_{pf} \vec{u} \cdot \nabla T + \nabla \cdot \{ [-\varphi \lambda_f + (\varphi - 1) \lambda_s] \nabla T \} = Q$ $(\rho_f c_{pf})_{ef} = (1 - \varphi) \rho_s c_{ps} + \varphi \rho_f c_{pf}$	Numerical simulation of a two-phase oil/water system. Pressure and temperature do not affect the boundary, and injection and production are balanced. The top and bottom boundaries of the model are no-flow and insulating, and the side boundaries of the model are set as constant pressure and open temperature boundaries.	[67]
Close-loop (Five-spot well pattern)	$Q_e = \frac{T_r - T_f}{\frac{1}{r_{re} h_o} + \frac{1}{2\pi} \sum_{i=1}^n \frac{1}{k_i} \ln \left(\frac{r_i}{r_{i-1}} \right)}$	The unsteady heat transfer model is based on a five-spot well pattern, and the thermophysical properties of the formation, casing, and cement sheath are independent of temperature and pressure. The materials are isotropic and homogeneous, and the fluid flow in the wellbore is one-dimensional, ignoring the radial temperature gradient in the wellbore.	[60]
Horizontal open loop	$Q_{wall} = \frac{2\pi(T_r - T_f)}{\frac{1}{r_o h_{int}} + \frac{\ln r_1}{2K_1} + \frac{\ln r_2}{2K_2}}$	The heat transfer in the GW system comprises convective and conductive heat transfer along the axis of the well, and the heat transfer between the surrounding rock column and the fluid to the GW axis. In this model, at the initial time, the pressure is based on the hydrostatic pressure. Heat transfer in fractures and different heat transfer performances are supplementary equations.	[68]
Close-loop (double-pipe heat exchanger)	$\frac{\partial T_{f0}}{\partial t} + \frac{\partial(vT_{f0})}{\partial z_v} = \frac{2(T_{fi} - T_{f0})}{\rho_{fluid} r_1^2 C_{fluid} \left(\frac{1}{h_1 r_1} + \frac{1}{\lambda_{insulation}} \ln \frac{r_2}{r_1} + \frac{1}{h_2 r_2} \right)} = S_r$ $\frac{\partial T_{fi}}{\partial t} + \frac{\partial(vT_{fi})}{\partial z_v} = S_r + \frac{2Rh_{wall}(T_{wall} - T_{fi})}{\rho_{fluid} C_{fluid} (R^2 - r_1^2)}$ $\rho_{rock} C_{rock} \frac{\partial T_S}{\partial t} = \frac{\partial}{\partial z_v} \left(\lambda_{rock} \frac{\partial T}{\partial z_v} \right) + \frac{1}{r_d} \frac{\partial}{\partial r_d} \left(r_d \lambda_{rock} \frac{\partial T_S}{\partial r_d} \right), R \leq r_d \leq r_\infty$ $h_{wall} (T_{wall} - T_{fi}) = \lambda_{rock} \left. \frac{\partial T_S}{\partial r_d} \right _{r_d=R}$	Neglecting the variation in tube wall temperature, heat exchange occurs between the fluid and rocks, and between injected and extracted fluids. This transient heat transfer model also includes the fluid flow equation, offering more completeness than previous studies, though it lacks field experiment verification.	[34]

The fluid output temperature from a closed-loop oil recovery well diminishes progressively during system operation, eventually reaching a state of equilibrium. Shortening the system stabilization time is achievable by lowering the fluid injection rate into the well, with minimal impact from the reservoir's heat capacity. The net power output relies on both the accumulated heat and the temperature of the fluid exiting the recovery well [69]. Sui et al. (2019) [70] illustrated that the exit temperature and overall energy markedly rise with enhanced fluid viscosity and thermal conductivity. Conversely, alterations in fluid density exhibit negligible impact on the potential quantity of geothermal heat extractable from the abandoned oil reservoir. In oil reservoirs containing geothermal reserves, permeability to interlayer interference plays a crucial role in the temperature and pressure differential of reservoir fluid production. Reservoirs with significant interlayer interference can achieve higher heat extraction more readily, making them a priority choice for heat extraction from AOWs [71].

A major challenge in the geothermal energy sector is the efficient transportation of hot water from GWs to surface thermal facilities. This process typically involves a pipeline network where heat loss is inevitable. Several factors influence heat loss, such as the temperature difference between the hot water inside the pipe and the surrounding environment, pipe specifications (e.g., diameter, length, material), insulation, and the duration of heat transfer. The production temperature of the hot water affects the extent of heat loss during fluid transfer.

In a system with a branched pipeline network for distributing heat from a thermal power plant to consumers, insulated pipes range from 65 mm to 250 mm in diameter, with a total pipeline length of 11,988 m. The average pressure in the primary network is 15 bar, with the hot water inlet and outlet temperature at 175 °C and 110 °C, respectively. In the secondary network, the temperature is 85 °C and 65 °C, and the pressure is at atmospheric levels. Table 4 provides estimates of heat and energy loss based on pipe diameter and length, calculated using thermal and exergy criteria, with energy loss determined through the thermal resistance of different materials. According to Table 5, heat losses in the pipeline network have been calculated using average daily external temperature values. The heat losses account for approximately 8.62 % of the total energy provided by the thermal power plant. This issue becomes even more pronounced when dealing with abandoned offshore oil wells, where elevated ocean currents and lower seabed temperature lead to significant heat dissipation during transportation. One common strategy to mitigate these losses involves insulating the seabed pipeline network, although this is a technically complex and costly approach. Alternatively, constructing a geothermal power plant on a decommissioned oil platform and converting geothermal energy into electricity or storing it as energy offers a more practical solution to counter these challenges [64].

Table 6 provides a comprehensive overview of previous studies evaluating open-loop (OLS) and closed-loop (CLS) systems for geothermal energy extraction from abandoned oil and gas wells (AOGWs). By analyzing key operational parameters, such as injection temperature, injection rate, well depth, tubing diameter, and geothermal gradient, Table 6 demonstrates their impact on production temperature and energy output, which may be utilized for electricity generation or direct heating applications. The 'Time' column indicates the operational duration (in years) from the start of geothermal energy production, offering insights into the temporal performance of these systems. This analysis is critical for identifying optimal well configurations that maximize energy efficiency, economic viability, and environmental sustainability. For instance, higher geothermal gradients and deeper wells, as shown in studies like [58,72], correlate with increased production temperatures (up to 182 °C) and energy outputs (up to 7900 kW), though they may require higher injection rates. Conversely, closed-loop systems, as seen in studies like [73,74], often prioritize sustainability by minimizing environmental impacts like fluid loss or subsurface disturbance, albeit with potentially lower energy outputs. By addressing gaps in feasibility assessments, Table 6 guides decision-makers in selecting cost-effective and sustainable configurations for repurposing AOGWs, supporting the transition to renewable energy while leveraging existing infrastructure to reduce capital costs. Table 7 focuses on key parameters that significantly impact energy extraction from GWs, particularly when AOWs are repurposed for geothermal energy. A thorough understanding of these factors is essential for the successful design, optimization, and implementation of geothermal systems.

4. Power production from abandoned wells

Although utilizing an abandoned well (AW) for electricity generation

Table 5
Heat and exergy loss according to the diameter and length of the pipe [64].

Pipe Diameter (mm)	Pipe Length (m)	Heat Loss ($\times 10^9$ kJ)	Exergy Loss ($\times 10^9$ kJ)
250	3227	11.032	3.067
200	1870	5.825	1.619
150	984	2.701	0.751
125	2759	7.008	1.948
100	1495	3.457	0.961
80	1393	3.092	0.859
60	260	0.528	0.147

may not initially seem ideal, increasing the wellbore's cross-sectional area and deepening the well can significantly enhance the heating surface area [37]. Converting repurposed oil wells into GWs for power production becomes more viable when multiple wells within an oil field are interconnected. This interconnected setup can collectively boost geothermal power output, despite the limited electricity generation capacity of individual wells due to low heat transfer rates in the surrounding rock formations [66]. For the management and energy efficiency of abandoned oil reservoirs, subsurface thermal energy storage (STES) technology is particularly considered during the summer season when energy demand is lower. In this scenario, seasonal heat storage allows for the recovery of stored heat to provide space heating during the winter months. It had been reported that at a depth of 200 m in an AOW, approximately 4.7×10^6 MJ of heat can be stored, with total heat extraction reaching around 2.9×10^6 MJ, which significantly exceeds the 8×10^5 MJ achievable without prior thermal storage advancements [59]. Given the lower efficiency of GE compared to fossil fuels, Noorollahi et al. (2017) [82] proposed combining solar energy with GE extracted from repurposed hydrocarbon wells to improve efficiency and reduce investment cost. This approach can help lower the fluid pumping cost associated with electricity generation. Another effective strategy for converting AWs into GWs involves using directional drilling or horizontal wells (Fig. 3) to increase their contact with high-temperature zones underground [47,83]. Implementing a triplet horizontal well configuration has been shown to be technically feasible and cost-effective for extracting low-enthalpy GE from depleted oil reservoirs [83].

Oil wells often encounter the challenge of excessive water production. To address this, Li et al. (2018) [84] proposed a multi-generation energy system driven by GE to extend the operational lifespan of high water-cut oil wells. Implementing such systems plays a crucial role in improving both energy efficiency and oil recovery during surface operations. In oil fields, enhanced oil recovery through in-situ combustion also enables the reservoir to reach high temperature. However, when steam flooding becomes economically unviable due to high water cuts or elevated steam-to-oil ratios, the reservoir is typically abandoned, leaving behind stored heat as potential energy. By converting AWs into heat exchanger systems, the opportunity to harness both thermal energy and GE arises. Fig. 8 illustrates the process of GE production using AWs combined with in-situ combustion. In this scenario, the AW is converted into an enhanced geothermal system (EGS) with a coaxial heat exchanger. The heat from the in-situ combustion process warms the GW, reducing the viscosity of the remaining oil and causing it to migrate toward the production well. As to the available reported data, the residual oil saturation in abandoned oil reservoirs often exceeds 30 % of the originally oil in place. According to existing studies, temperature in these reservoirs can exceed 400 °C in light oil formations and reach over 600 °C in heavy oil formations through thermal processes such as in-situ combustion techniques. Under such high-temperature conditions, these oil reservoirs may transform into highly advanced geothermal reservoirs, offering superior performance compared to conventional advanced geothermal systems in hot dry rock formations [85,86]. According to established standards, water extracted from oil fields is typically classified as either medium-enthalpy (90 °C to 150 °C) or low-enthalpy (30 °C to 90 °C) geothermal resources [28]. For electricity generation, a process involving steam injection followed by water injection can produce high-temperature, high-pressure steam. The efficiency of electricity production from in-situ combustion reservoir fluids exceeds that of conventional hot fluids extracted from hydrocarbon reservoirs, primarily due to the higher temperature achieved. An additional advantage is that the extra crude oil production also helps extend the operational lifespan of the reservoir, thus supporting the oil industry for a longer period [86].

Table 6

Summary of previous studies on open-loop and closed-loop well systems showing impacts of key parameters (e.g., injection rate/temperature, well depth/diameter, geothermal gradient) on production temperature and energy output over operational time.

production temperature (°C)	Injection temperature (°C)	Generating electricity or direct heating (kW)	Well system	Injection rate ($\frac{m^3}{h}$)	well depth (m)	Inside diameter of the tubing (m)	Geothermal gradient (°C/Km)	Time (year)	Reference
125.5–126.7	–	400–500	Open-loop	–	6100	0.088	28	3.7	[75]
114.6	80	400–700	Close-loop	32	400	0.24	34	0.5	[73]
138	30	947	Close-loop	15	4000	0.05	45	0.16	[58]
59	50	627.9	Close-loop	10.8	3000	0.076	–	–	[65]
38	30	345	Close-loop	36	3500	0.0381	36	5	[74]
28	10	170	Close-loop	8.33	2500	0.031	33	0.41	[22]
182	30	7900	Open-loop	180	4500	–	43	30	[72]
68–72	50	8000–1200	Open-loop	360	5500	–	21	20	[76]
70	50	3500	Open-loop	576	2000	0.1397	30	10	[67]
85–90	110	310	Open-loop	120	3216	–	35	–	[77]
64.3	20	–	Close-loop	7.2	3861	0.12	31	–	[23]

5. Techno-economic evaluation of converting abandoned wells into geothermal wells

When converting AOGWs into GWs, specialized enhancement techniques and geological uncertainties play a crucial role. These factors are particularly important during the exploration and development stages, as they can significantly reduce project uncertainty when evaluating the methods and feasibility of investment decisions for predicting GE extraction [17,58]. Despite its potential, the repurposing process is accompanied by significant technical challenges and economic considerations. The following sections provide a detailed examination of these aspects, focusing on the practical limitations (Section 4–1) and the associated cost (Section 4–2) of converting abandoned wells into geothermal systems.

5–1- Practical challenges and limitations in repurposing abandoned wells for geothermal energy extraction.

Evaluating the feasibility of converting a repurposed well into a GW requires careful consideration of both technical and economic factors. The viability of repurposing oil and gas wells for GE production, along with the associated challenges, must be thoroughly assessed. Additionally, technical, market, and policy barriers that may hinder the advancement of geothermal energy projects require comprehensive evaluation. The selected well must exhibit geothermal resource potential such as adequate depth, a favorable geothermal gradient, and stable temperature output over a specified period as well as conditions suitable for reinjection [87,88]. Since nearly half of production wells have been repurposed due to casing issues [22], maintaining the production capacity of GWs presents challenges. One such challenge is adapting wells that were originally designed according to oil and gas standards to geothermal environments. For example, an assessment of the technical feasibility of reactivating wells in Indonesia highlights the necessity of establishing a comprehensive well assessment methodology. This includes casing design, historical data analysis, identification of well-related problems, and evaluation of well integrity to address the challenges associated with repurposing wells for geothermal use [88]. Drilling high-temperature wells for geothermal applications presents several distinct challenges compared to drilling in the oil and gas industry. Over the coming decades, we anticipate significant technical hurdles associated with varying temperature levels across the globe. In addition to temperature, several other critical differences between

drilling wells for oil and gas production versus geothermal heat extraction should be considered:

1. GWs require drilling through harder and more abrasive formations, such as volcanic and permeable sedimentary rocks.
2. Corrosive salts present in reservoir fluids during GW drilling.
3. Typically, enlarging well sizes to maximize flow during heat production [6,89].

Abandoning oil and gas wells is more challenging and costly than the initial exploration phase. The process involves developing a detailed safety and engineering abandonment plan, ceasing extraction, sealing the wells, complying with regulatory requirements for structure removal, and managing material disposal or recycling. The associated cost depends on factors such as the location, well complexity, site characteristics, and weather conditions [89]. Another challenge in geothermal systems, particularly with open-loop GWs, arises from the interaction between injected fluids and reservoir fluids, leading to potential scaling issues. This necessitates careful well completion decisions to mitigate such risks [44]. Well abandonment is a complex operation that involves various risks, including those related to analysis, facility removal, operational procedures, waste management, and regulatory oversight. Precise and thorough preparation is essential for the successful execution of well plugging and abandonment, as improper abandonment can be both technically challenging and financially costly. The conversion of an oil or gas well into a GW introduces additional challenges. During the drilling phase, obstacles include the management of drilling fluids, which pose pollution and environmental risks, and casing, which can result in fluid leaks and reduced well integrity. Wellhead equipment must also be adapted to withstand the higher temperature and pressure associated with geothermal wells, and blowout preventers used in GWs differ from those in conventional wells. Furthermore, during the production phase, several issues arise, including the erosion of casing cement, which can lead to leaks, casing corrosion, and the need to replace or modify components such as packers for new conditions. Tubing must be capable of withstanding the elevated temperature and pressure typical in geothermal operations, and additional components like the downhole safety valve (DSV), tubing hanger, and wellhead must be designed to handle the new operational loads. The wellhead originally designed for oil or gas extraction may not

Table 7
Key parameters for converting AOWs to GE systems.

Factors	Main Observations and Requirements	Ref.
Pressure	Injecting fluid at a moderate pressure allows it to heat up and produce steam, ultimately maximizing the system's performance, which depends on the injection pressure. For a depth of 3000 (m) with a temperature of 450 (K) in a closed-loop wellbore, an injection pressure of 30 (bar) is optimal.	[78]
Horizontal Hole	The temperature of the water returning from horizontal wells is influenced by the length of the horizontal section used for heat transfer. However, this effect stabilizes after approximately 1000 m, indicating that this length is sufficient for effective heat transfer from the geothermal zone to the injected water.	[79]
Pipe Insulation	The temperature of the returning water decreases rapidly as the thermal conductivity of the pipe insulation increases within the low thermal conductivity range.	[79]
Inlet Water Temperature	Increasing the inlet water temperature from 5 °C to 25 °C reduces the number of wells suitable for low-grade heating by two-thirds, with minimal impact on wells with higher outlet temperature. Additionally, every 5 °C increase reduces the heat transfer rate by an average of 23 %.	[41]
Working Fluid	Increasing viscosity raises the outlet temperature and total energy, but the fluid's density has little effect on the amount of extractable heat. The specific heat capacity of the fluid can increase the total thermal energy produced, but it may reduce the returning fluid's temperature.	[71,80]
Reservoir Lithology	Reservoir lithology, such as sandstone, carbonate, and crystalline rocks, requires specific EGS technologies and hydraulic fracturing techniques for efficient energy extraction. Each rock type has unique characteristics that influence the choice of technology and methods used to maximize GE production.	[30]
Depth	The energy grows linearly as the depth of the borehole increases.	[81]
Well Spacing	Pressure and flow changes usually happen within 50 m of the wellbore, while the cooling effect (rock or fluid temperature around the well) can reach up to 85 m. This distance is important for planning the fracturing area and spacing wells when converting an AOW to a geothermal system.	[80]
Well Selection	In OLSs, the water's pressure, temperature, and flow rate depend on factors like the depth of the well, the geothermal gradient, and the porosity and permeability of the reservoir rocks. In CLSSs, in addition to focusing on temperature, it's essential that the annulus is fully cemented.	[38]
Injection Flow Rate	As the flow rate increases, the temperature drops, but power output rises. However, higher flow rates lead to higher operational cost due to increased pump energy use and maintenance. Therefore, optimizing the flow rate is essential for balancing efficiency and cost.	[81]

be suitable for the extreme conditions in a geothermal well and may need to be replaced.

During the abandonment phase, components such as the wellhead and tubing are retrieved from the well, and wellbore sections that may allow fluid migration are securely plugged and cemented to ensure proper isolation and prevent environmental hazards. Cement plugs are placed inside the wellbore to isolate different zones, regardless of whether the tubing is left in place or removed. The wellhead is then removed, ensuring that all sections of the well are safely sealed to prevent fluid migration and environmental risks [38]. Fig. 9 illustrates a schematic of the well lifecycle and associated barriers. In the drilling section, primary barriers include drilling fluid, while secondary barriers consist of the blowout preventer (BOP) and casing. In the production

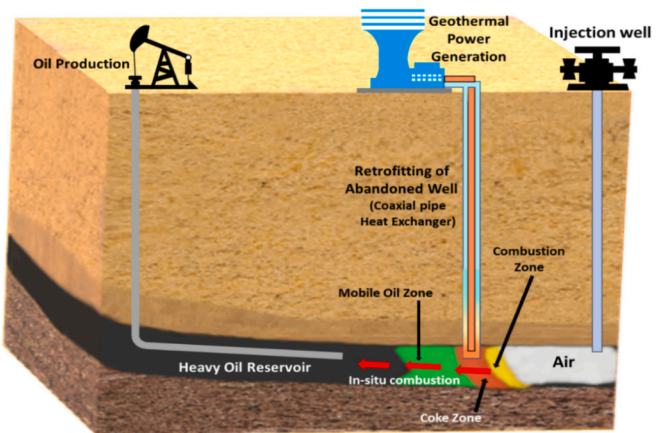


Fig. 8. Generating GE through repurposed oil wells and in-situ combustion method.

section, primary barriers include the downhole safety valve (DHSV) and wellhead (X-mas tree), while secondary barriers consist of casing and cement. The intervention section involves maintenance operations to preserve or enhance well performance, while the plug and abandonment section entails placing cement plugs as primary and secondary barriers to isolate different zones and prevent fluid migration [90]. To abandon a well, it must be restored to its pre-extraction state, a process that presents several challenges. The decision to repurpose a hydrocarbon well into a GW should be made in advance or during a period when the well is not in operation. This proactive approach helps to prevent additional costs associated with changes, repairs, and maintaining well integrity. It also facilitates better equipment and design planning, as well as more effective financial and time management, all while addressing environmental concerns.

Abandoning offshore wells is more complex than onshore wells due to key parameters such as well depth and well pressure. For offshore wells, additional factors like the type of facilities, serviceability, water depth, and well pressure must be carefully considered. These factors not only complicate the abandonment process but also influence the cost involved in completing it. Recording operational data during oil and gas well operations is vital for transforming these wells into geothermal wells. However, it is important to recognize that many of these wells are located in regions with limited GE potential. As a result, these wells typically exhibit lower to moderate temperature. Furthermore, many of these wells lack sufficient subterranean resources, which means that fluid injection into the wells is unnecessary [92].

5.1. Cost analysis of converting abandoned wells into geothermal wells

The utilization of AOGWs, which have previously lost their economic value, for GWs results in time and drilling cost savings. This, in turn, leads to a reduction in the Rate of Return on the investment made and reduces the initial drilling cost of AOGWs by 40 % to 60 % [93]. From an economic perspective, a critical factor determining the feasibility of extraction is the amount of extracted GE, as it directly influences the total project cost, including capital, operational, and other expenses. Assessing the potential benefits of geothermal power plants against this cost is essential before initiating a project. Fig. 10 illustrates and discusses the cost associated with well abandonment, conventional GW development, and the repurposing of AOGWs into GWs [30]. The retrofitting of a suspended well is, on average, \$50,000 less expensive than retrofitting an equivalent AW. The cost for retrofitting suspended wells remains relatively consistent, typically falling below \$11,000. Table 8 offers a detailed breakdown of the extra expenses associated with retrofitting an AW.

Furthermore, Table 9 presents a categorized list of fixed costs

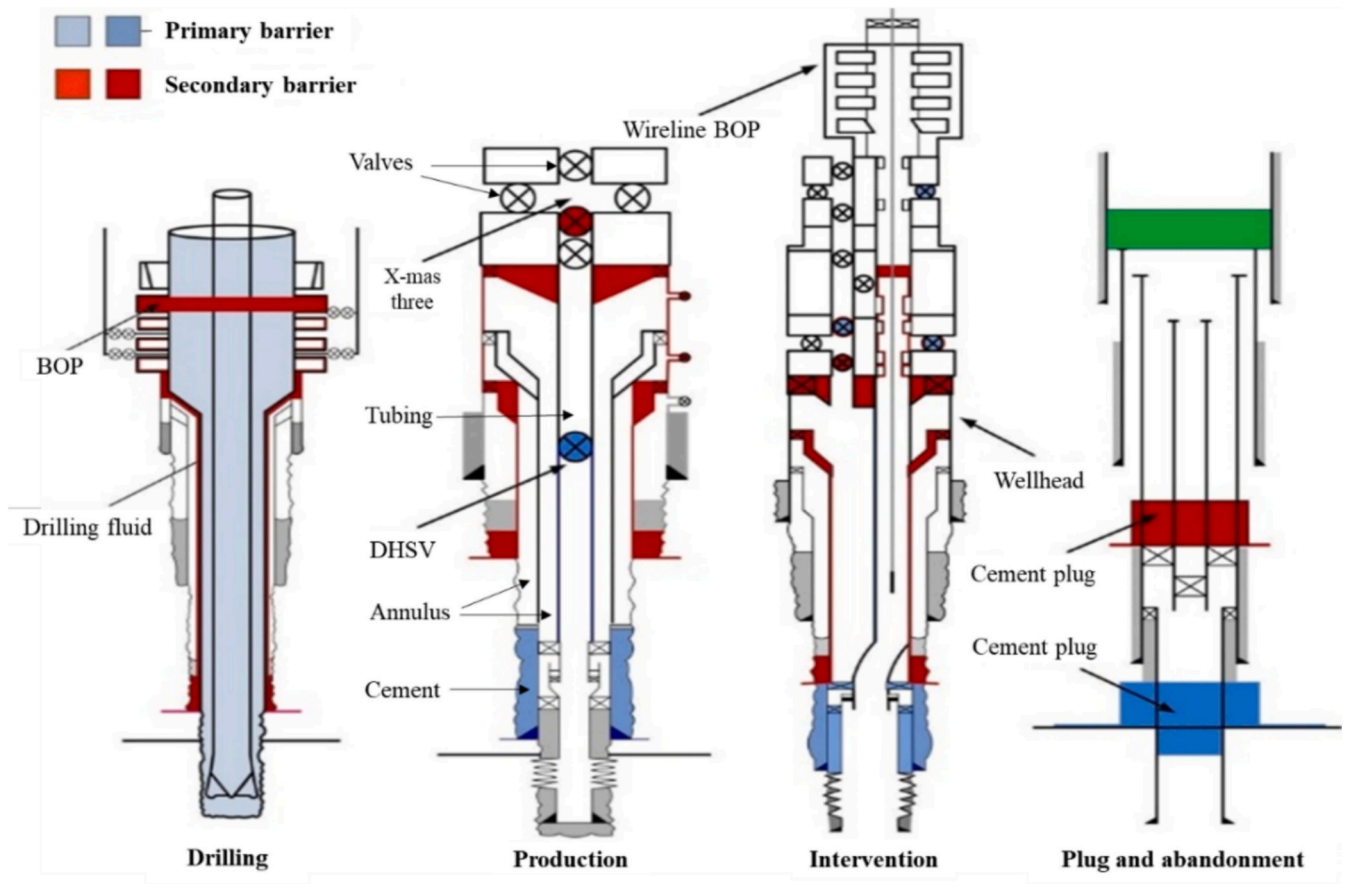


Fig. 9. Simplified schematic of well barriers for well abandonment (modify after) [91].

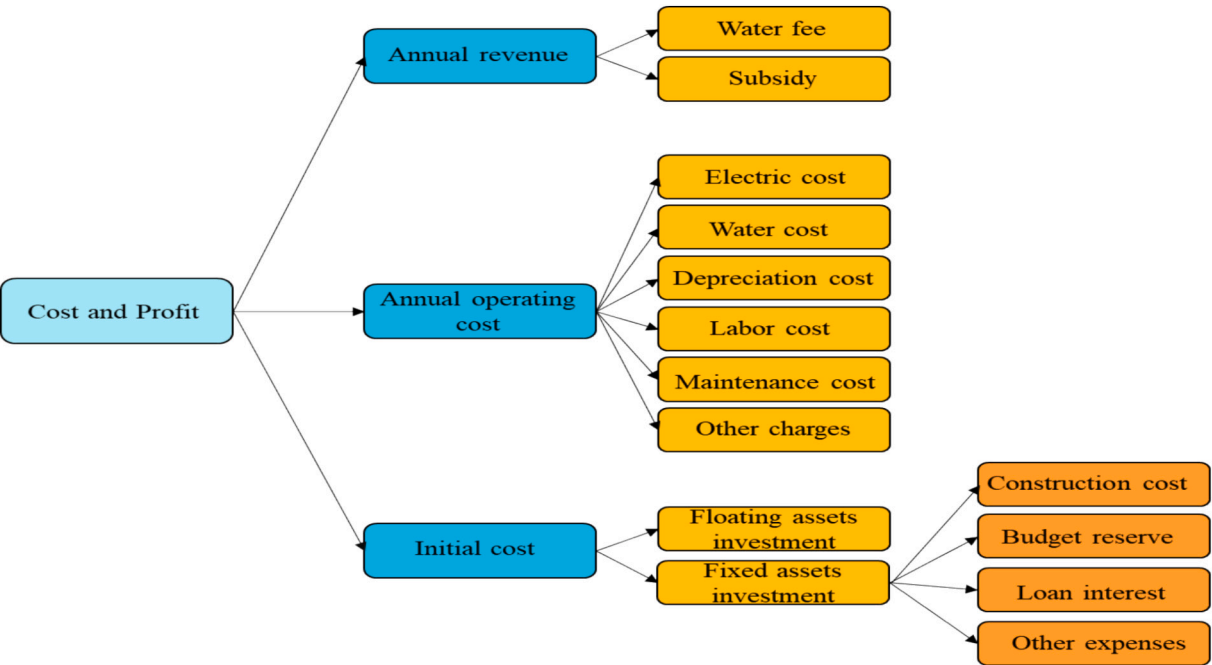


Fig. 10. Cost and profit in GE production (modify after) [30].

associated with retrofitting a hydrocarbon well for GE production. Both Tables 8 and 9 are based on a study by the Petroleum Services Association of Canada (PSAC), which forecasts the prices of oilfield services,

equipment, and materials in Western Canada. These prices are derived from the updated Winter PSAC Well Cost study and are used to estimate the cost of retrofitting wells in the foothills region of Alberta for GE

Table 8

A comprehensive list detailing the additional cost associated with refurbishing an AW. It is important to highlight that certain cost components, denoted with an asterisk (*), are calculated daily and are impacted by the duration of service rig usage [94].

Additional Expenses for a Decommissioned Well	Expense Per Day/Unit (\$)	Overview
Cut & cap replace	870	10 m of 244.5 mm 53.57 kg/m J-55
Welding	1500	Day rate
Service Rig*	7750	Rig and Crew
Transportation*	2500	Hauling of equipment
Other Services*	4500	Vacuum truck
Completion Fluids*	4500	Water + trucking
Logging (cement bond log)	4250	Additional CBL (check old cementing)
Wellsite Supervision*	1400	Extra days of site supervision
Environmental*	1000	Environmental technician per day
Lease & Road Maintenance*	500	Service days
Miscellaneous Cost*	–	8 % Cost overrun buffer
Overhead*	–	2 % Administrative addition

Table 9

Fix cost related to converting oil/gas wells for GE generation [94].

Planning and Lease Preparation	Item Expense (\$)	Overview
Well License/Applications	500	Regulatory permit
Preparation & Roads	8700	1 day of lease cleanup + supervision
In-house Engineering (Drilling)	16,000	100 h of engineering work
Equipment Inspection	3500	Casing inspection log
Misc. Cost	2296	8 % Cost overrun buffer
Overhead	574	2 % Administrative addition
Subtotal	31,570	–
Well Equipment		
Packer	12,750	To seal wellbore from oil or gas producing zone
Wellhead	18,000	To replace old wellhead
Misc. Cost	2460	8 % Cost overrun buffer
Overhead	615	2 % Administrative addition
Subtotal	33,825	–

production [94]. With variations based on the well's depth and scenario (open well, cased and screened, or completed), the expenses can range from 22,855 to 164,673 Canadian dollars [95]. By adopting optimization, abandonment costs can be minimized each year, which are reported to be between \$52,000 and \$76,000 [96]. Below the surface, cost may include the removal of the drilling rig, while on the surface, they encompass site cleanup for each well, removal of cables and pipelines, and from an environmental perspective, the removal of contaminated soil. Given the substantial expenses involved in well abandonment, repurposing a well for alternative uses, such as GE extraction, presents an appealing option. This not only reduces costs for the company but also extends the economic life of the well. However, such retrofitting should be strategically planned during the initial preparation phase. For wells that are permanently closed, adaptation becomes impractical and significantly more challenging, especially when the well depth is insufficient, necessitating additional drilling [30]. The expenses related to the exploration and establishment of a GW can be divided into two main components. The first relates to the investment in surface equipment and systems, while the second pertains to subsurface investment, including the drilling of GWs. As a result, a considerable portion of the drilling cost for GWs can be reduced by utilizing AOGWs. Also, It's important to mention that deeper wells tend to incur higher start-up cost

[30].

It has been reported that the extraction and re-injection of working fluid in a geothermal reservoir can account for up to 50 % of the total cost, ranging from 500 to 4000 (\$/kW) produced. However, converting repurposed oil and gas wells into GWs can reduce overall cost by up to 50 %, primarily due to the elimination of drilling and decommissioning expenses. Additionally, the construction of the power plant constitutes about 40 % of the total cost, equivalent to 1500 to 1700 (\$/kW) produced, resulting in a net cost of approximately 2000 to 6000 (\$/kW). In contrast, operation and maintenance account for about 10 to 20 % of the net cost. Finally, retrofitting and optimizing an AOGW to function as a GW is regarded as a more cost-effective and sustainable alternative to decommissioning the well [97].

6. Environmental impact of repurposing abandoned wells into geothermal wells

When oil and gas wells are not properly abandoned, the migration of fluids into upper layers can result in serious environmental damage [26]. Decommissioning the well and leaving it unused requires careful consideration, as improper handling of AOGWs may pose safety hazards, lead to leakage, and cause environmental damage [98]. The dangers caused by an AOGW can affect up to 95 % of the environment around the well [99]. A well that has not been correctly sealed will inevitably present risks with possibly catastrophic consequences. Even in cases where wells have been properly sealed, complications can still arise. Escaping liquids and gases from AOWs can severely harm both the environment and human health. These wells can become conduits for methane and other subsurface gases to enter the atmosphere. Additionally, they are often linked to groundwater contamination. Various factors, such as geological conditions, well age, shallow gas presence, and adherence to best practices and regulations, influence fluid leakage in abandoned wells.

From an environmental perspective, GE use offers more advantages than traditional oil and gas extraction. The levels of gases like carbon dioxide in AOGWs converted to GWs are almost zero, and for open-loop GWs, they are negligible [30,100]. Using GE reduces the environmental cost of CO₂ emissions by approximately \$19.13 million per year, while without the use of GE, it amounts to \$30.98 million [100]. While GE is considered a renewable and clean energy source, only small amounts of greenhouse gases are found in geothermal reservoirs. Using CLSs in AOWs for GE production reduces the production of these fluids to zero. In Table 10, the emission of gases in various power plants is mentioned, where the emission contribution in closed-loop geothermal power plants is zero, in contrast to the significant emissions from hydrocarbon fluids. This fact is environmentally significant [101]. The high amount of 17.6

Table 10

Gas emissions from different power plants [101].

Plant type	CO ₂ Kg/MWh	SO ₂ kg/MWh	NO _x kg/MWh	Particulates kg/MWh
Coal-fired	994	4.71	1.955	1.012
Oil – fired	758	5.44	1.814	N. A ¹
Gas – fired	550	0.0998	1.343	0.0635
Geothermal - Flash Steam, Liquid Dominated – USA	27.2	0.1588	0	0
Geothermal – The Geysers dry steam field – USA	40.3	0.000098	0.000458	Negligible
Geothermal – Closed loop binary/EGS	0	0	0	Negligible
Geothermal – Flash steam – Hellisheidi – Iceland	21.6	17.6	0	0
Geothermal – Flash steam – Tuscany – Italy	324	1.65	–	–
Average, All European plants	369.7	1.1	0.5	0.1

¹ Not Available

kg/Mw.h SO₂ in the Hellisheidi power plant is primarily due to the unique chemical composition of the geothermal fluids in the region and the associated production and control processes. In the flash steam process, geothermal fluids consisting of steam and hot water are extracted from the ground. If these fluids contain high amounts of H₂S, during the process, H₂S is converted to SO₂. Table 11 refers to studies on reducing carbon dioxide emissions through the utilization of AOGWs for GE. CO₂ storage, also achieved through the use of AWs, holds promising prospects for pollution reduction and GE production in the future.

Repurposing AOGWs for GE extraction significantly reduces environmental impacts compared to traditional oil and gas operations, with closed-loop systems achieving near-zero greenhouse gas emissions and substantial CO₂ reductions (e.g., up to 215 million metric tons/year in the Williston Basin [102]). The proper well sealing is critical to prevent methane leakage and groundwater contamination, which can affect up to 95 % of the surrounding environment [99]. Integrating technologies

Table 11
Summary of previous study on pollution reduction using AWs for GE.

Main conclusion	Reduction CO ₂	Field (Country)	Ref.
The direct use of AWs results in a Levelized Cost of Heat (LCOH) equal to \$12.0/MMBTU, and the heat generated by the well amounts to 1,025,602,926,446.69 BTU/year.	215 million metric tons/year of CO ₂	Williston Basin (USA)	[102]
Using repurposed oil wells for 50 % of a desalination plant's freshwater capacity significantly reduces CO ₂ emissions from 17,842,587 to 1,287,058 tons, and decreases NO _x and SO _x emissions from 25,416 to 6294 tons and 36,640–22,343 tons, respectively.	16.5 million tons	Ahwaz oil field (Iran)	[103]
Generating electricity from an oil well using water circulation in an Organic Rankine Cycle (ORC) unit, with a maximum electrical power capacity of 110 kW, at a water flow rate of 22.1 l/s and an outlet water temperature of 122 °C, falls within the LCOE range of \$0.10/kWh to \$0.54/kWh.	8600–14,800 tons	Oil well, Alberta (Canada)	[104]
An optimized model of AOGWs in Forest Reserve fields, equipped with hydraulic fractures, comprises 3 injector wells and 3 producer wells. The current net worth of this model is \$1,431,263,840, with a geothermal ground electricity cost of \$0.05/kWh and a cumulative enthalpy of 2.8E+12 Btu.	50,062,500 tons/25 year	South-western (Trinidad)	[105]
The combination of CCS (Carbon capture and storage) technology with GE production using CO ₂ as the working fluid in the depleted oil well resulted in 47,004 tons of CO ₂ being stored per day and the production of 1.997×10^{16} J of energy at an injection pressure of 20,000 kPa.	1,984,000 tons	Forest Reserve Field (Trinidad)	[106]
Utilizing AOWs for geothermal heating demonstrated efficient heat transfer at a depth of 3000 m, producing 5.5×10^{12} J of GE. This approach saved 185 tons of coal and reduced annual CO ₂ emissions by approximately 678 tons.	457 tons/ year	Shengli oil field (China)	[107]

like CO₂ storage enhances sustainability, making AOGW repurposing a vital strategy for clean energy production and ecological preservation. Future efforts should focus on advancing sealing techniques and scaling CCS applications to maximize environmental benefits.

7. Combining abandoned wells with geothermal resources for simultaneous utilization

The co-generation performance of heat and oil is superior, emphasizing the significant impact of the oil phase in the well patterns system during oilfield geothermal production [67]. Given that a considerable number of oil wells worldwide have been abandoned or will be in the future, a substantial amount of oil and energy remains trapped within the rocks in these abandoned reservoirs due to technical and economic constraints [108]. Simultaneously generating GE and hydrocarbons can be a viable solution both economically, in terms of energy production, and environmentally for reducing zone pollution and revitalizing AWs. GE sources in oil fields can be tapped through two avenues:

- (1) **Utilizing high-temperature co-produced water** from deep hydrocarbon wells or from high-temperature shallow formations. For example, at the Wyth Farm Oil Field in the UK, a geothermal project has been implemented to harness GE from co-produced water.
- (2) **In-situ combustion technique** within the reservoir, typically employed for hydrocarbon recovery, can also be used to harness geothermal energy from repurposed oil wells [109].

Fig. 11 illustrates the evolution of a hydrocarbon field into a geothermal field for GE production. It starts with identifying suitable candidates, characterized by wells with a water cut of at least 50 % and wellhead temperature surpassing 90 °C. The subsequent joint production project assesses the energy production potential, followed by the commissioning phase. The ultimate goal is to convert the hydrocarbon field into a geothermal field. Achieving this involves a dedicated conversion project, focusing on dismantling hydrocarbon facilities and securing essential permits [75].

A comprehensive checklist for assessing hydrocarbon fields with geothermal objectives is presented in Fig. 12, encompassing stability for all identified aspects affecting the utilization of GE [13]. Zhang et al. (2023) [110] demonstrated that the income distribution for repurposing oil fields comprises 54 % from CO₂ enhanced oil recovery, 40 % from CO₂ storage, and 6 % from geothermal development. Conversely, in repurposing gas fields, the revenue distribution is 12 % from CO₂ enhanced gas recovery, 82 % from CO₂ storage, and 6 % from geothermal development. Notably, transforming wells to enhanced hydrocarbon recovery, CO₂ storage, and geothermal development yields 18 % higher revenue in oil fields compared to gas fields. In heavy oil reservoirs, in-situ combustion is often utilized as a thermal method to enhance recovery, particularly when the formation of hydrates is present. This technique not only elevates rock temperature to promote hydrate formation but also provides an opportunity to harness GE from the reservoir. One of the challenges leading to well abandonment in gas wells is the formation of hydrates around the wellbore due to changes in temperature and pressure, which can result in wellbore blockage. The cost associated with well plugging and damage caused by hydrates can be significant, reaching up to \$30 million per incident and exceeding \$100 million in offshore operations [111,112]. But, since natural gas hydrates (NGH) have recently emerged as a potential alternative energy source, a series of innovative approaches has been proposed for offshore NGH development, combining depressurization and thermal flooding by utilizing geothermal heat from existing hydrocarbon wells through a new multilateral well system (Fig. 13). In this system, a vertical primary well is drilled into a deep oil and gas reservoir to generate thermal fluid. Lateral wellbores, positioned above and below the shallow NGH reservoir, extend from the vertical well to tap into the NGH reservoir

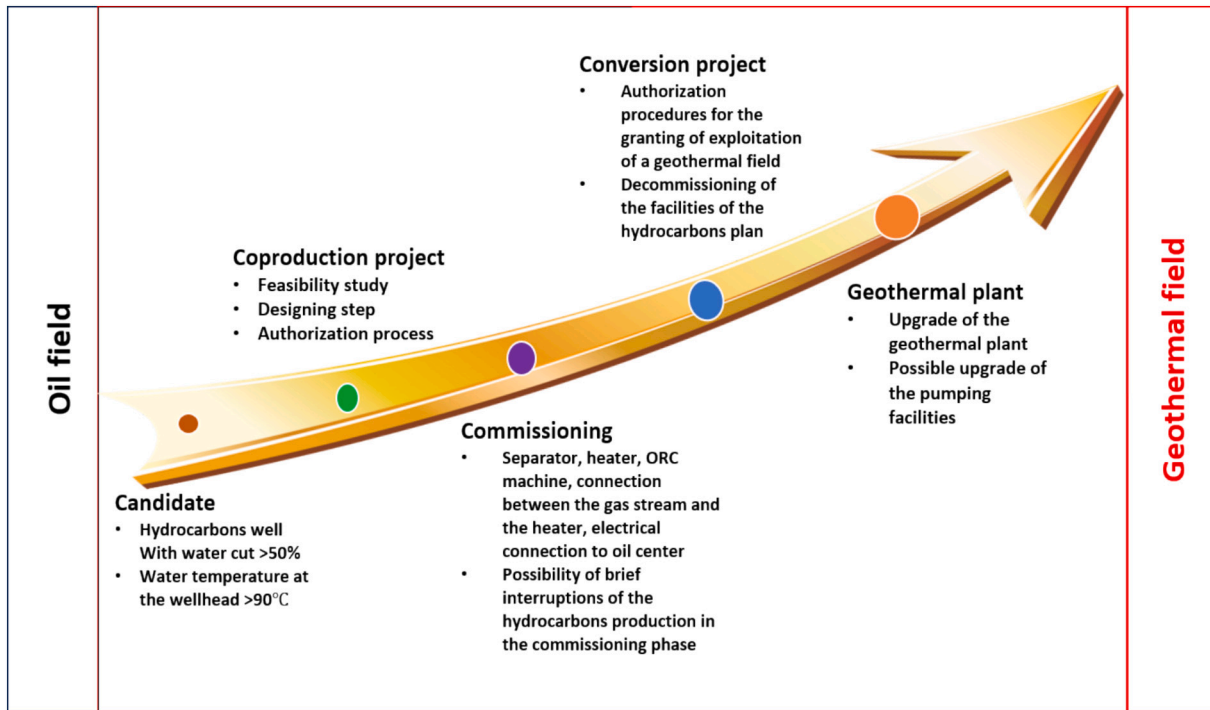


Fig. 11. The stages of transforming an oil field into a geothermal field (modify after) [75].

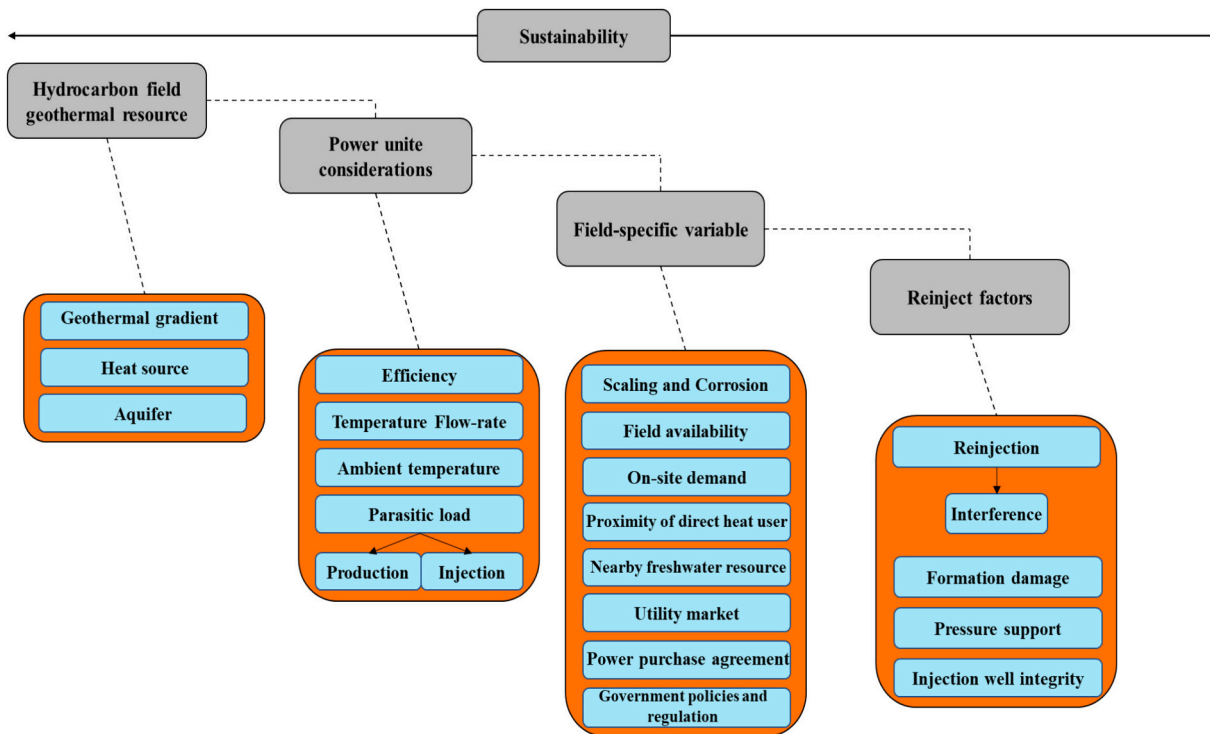


Fig. 12. Factors that impact the utilization of GE in hydrocarbon fields at different levels (modify after) [13].

independently. Upper and lower packers are installed within the multilateral wells system, allowing geothermal energy to be transferred into the hydrate-bearing zone, causing the dissociation of hydrates. As a result, oil and gas are continuously produced while exploiting the NGH reservoir. This multilateral well system offers a cost-effective solution for offshore NGH development by reusing existing oil and gas platforms, which eliminates the need for additional expensive infrastructure like

floating production platforms, subsea wellheads, and other related drilling systems. This approach not only reduces construction costs but also enhances operational safety and improves production maintenance. Field data from hydrocarbon fields can be employed to evaluate the feasibility of NGH development. The use of fewer wells in offshore NGH exploration reduces geological and engineering risks compared to conventional oil and gas well operations while avoiding well abandonment

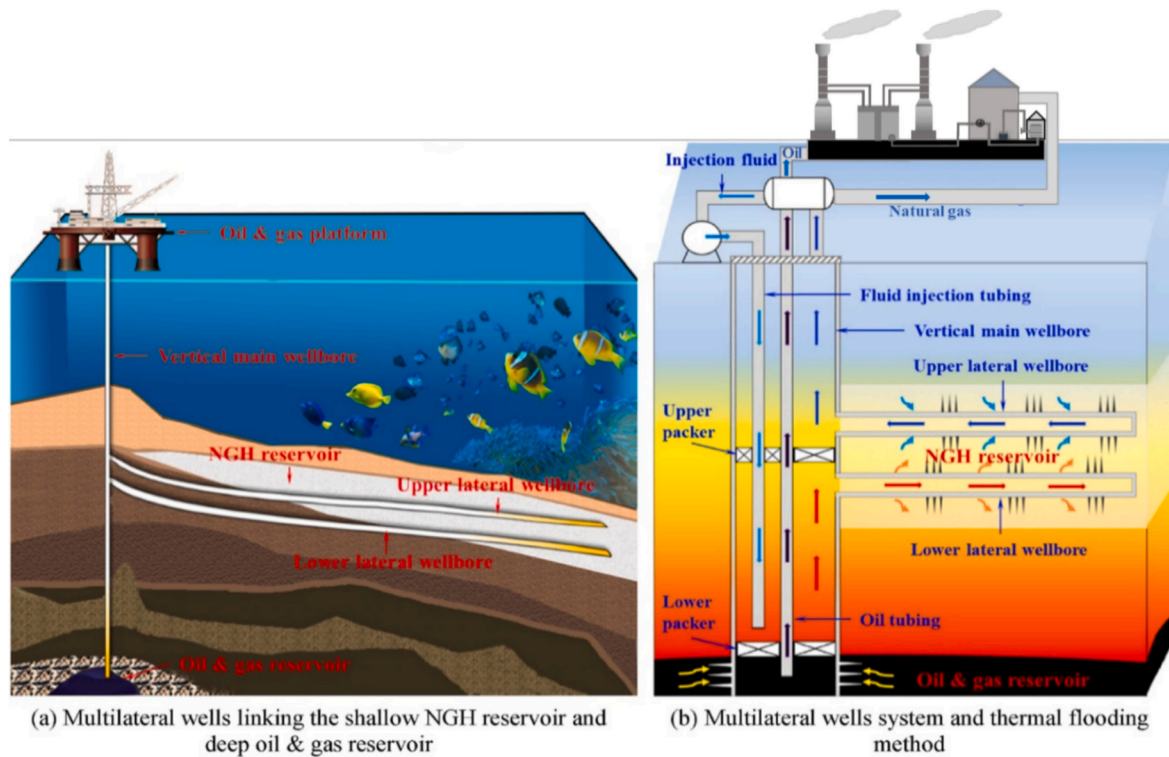


Fig. 13. A schematic of the multilateral well system and thermal flooding approach in natural gas hydrate (NGH) and oil extraction (modify after) [112].

contributes to higher natural gas production rates. Studies estimate that this method can recover approximately $1.6 \times 10^7 m^3$ of natural gas in one year [112].

The transition of repurposed oil and gas reservoirs to exceptional enhanced geothermal reservoirs, where the remaining oil and gas saturation is typically over 40 %, can be significant [113]. The abandonment of oil wells typically arises from economic constraints and declining production efficiency, often due to excessive water production or pressure reduction. However, in such scenarios, it is possible to extend the productive life of these wells by injecting fluids to maintain reservoir pressure. This approach allows for the repurposing of hydrocarbon wells into GWs, effectively diverting the water cut from the hydrocarbon wells toward GE production. Additionally, this method offers the potential to recover the remaining hydrocarbons in the reservoir, maximizing the utility of both energy sources. Additionally, drilling more AWs can help access zones with a suitable thermal gradient. In such cases, by separating the produced water from hydrocarbons and ensuring minimal injection of aquifer water into the oil and gas production wells, the separated water can be reinjected into the GWs. However, it's crucial to estimate the separation cost in this process. Regarding pressure reduction, injecting water from GWs can compensate for pressure deficiencies in hydrocarbon reservoirs. However, alternative methods might be required for reservoirs experiencing pressure decline due to hydrocarbon extraction. Also, it's important to assess whether the injected water impacts the aquifer zone or not.

The simultaneous production of oil and GE presents a novel approach in the energy industry. To implement this, the first step is to evaluate an oil or gas reservoir. A water injection system is then established in the subsurface reservoir to generate GE. The injected water must have a sufficiently high temperature and pressure to generate GE. Once injected, the water produces steam, which is extracted and converted into electricity via steam turbines. If the water injection process is correctly managed, this method could also enhance oil and gas production.

In the OLS, injected water moves upward from the lower layers. This water helps maintain reservoir pressure and sweeps oil toward the production wells, improving sweep efficiency. At the same time, the

water absorbs geothermal heat, which can be utilized for GE production. However, for the simultaneous production of oil and GE, advanced technologies and the optimal use of both water and energy resources are essential. This approach may face challenges such as technical complexity, engineering issues, and high cost in retrofitting wells, requiring a detailed economic and engineering evaluation of the project. The well design method for the simultaneous production of hydrocarbons and geothermal energy is illustrated in Figs. 14 and 15. Figs. 14 and 15 show (a) the simultaneous production of oil and GWs in an OLS through the creation of hydraulic fractures, and (b) a CLS. It is important to note that integrating GWs into AWs with smaller casing diameters may present technical challenges. These challenges can be mitigated by establishing the conditions for geothermal well creation during the

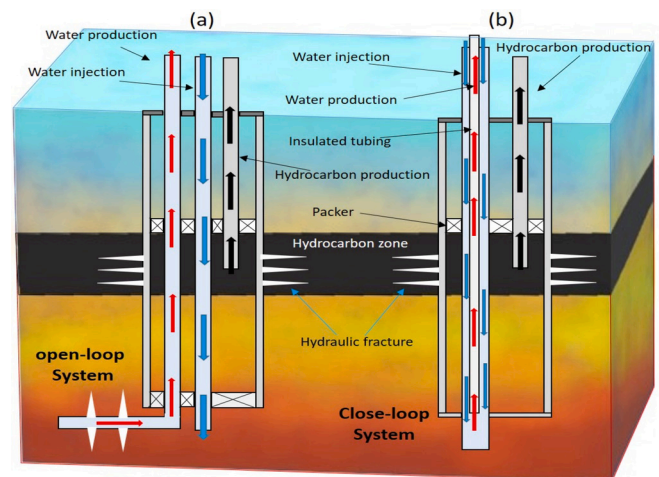


Fig. 14. A schematic of the separated pipe system for simultaneous hydrocarbon production from an AW and GE is illustrated as follows: (a) a horizontal open-loop system (OLS) with hydraulic fractures, and (b) a close-loop system (CLS).

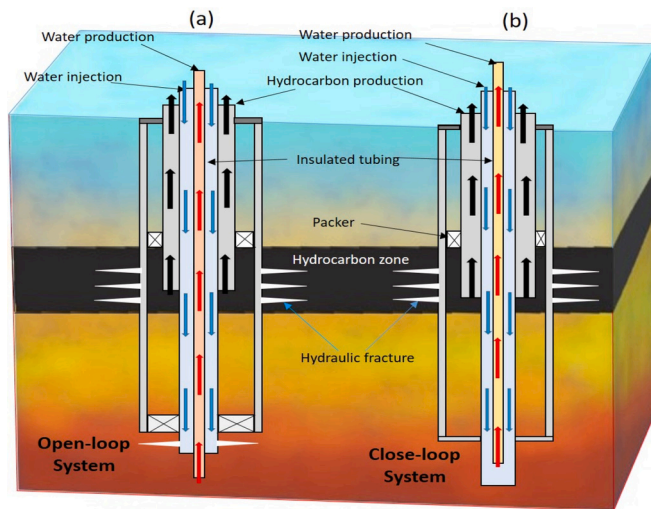


Fig. 15. A schematic of the centralized system for simultaneous hydrocarbon production from an AOW and GE: (a) an open-loop system (OLS) with hydraulic fractures, and (b) a close-loop system (CLS).

initial drilling phase of oil and gas wells, prior to abandonment, or at the onset of simultaneous production with hydrocarbons from existing reservoirs. Early planning and design adjustments are crucial to overcoming such limitations.

For effective heat transfer during fluid circulation at the bottom-hole, sufficient contact area with the reservoir is essential. To achieve high surface temperature, an appropriate energy rate, and high fluid enthalpy, an OLS combined with horizontal well drilling (Fig. 14b) is a suitable option. However, since the fluid is in direct contact with the reservoir, OLS presents more challenges compared to a CLS. Despite these challenges, OLS can offer better conditions for electricity generation due to its higher heat recovery potential.

8. Projects and economic studies in abandoned oil and gas wells

While some studies have explored the economic aspects of AOWs² into GWs, comprehensive analyses of the overall economic impact remain scarce. Limited evidence suggests that the expenses associated with establishing a new GW might be lower than the cost involved in decommissioning an oil well. Understanding the factors influencing these outcomes requires in-depth scrutiny to foster greater advancements in extracting energy from AOWs [58,114]. One of the key differences between GWs and oil wells is the production rate, with oil wells typically yielding much less than GWs. Therefore, it is evident that utilizing the facilities and infrastructure already present in oil fields can significantly reduce the initial cost of geothermal projects and make them economically viable [114]. Nian and Cheng (2018) [107] reported that the annual cost of utilizing geothermal heating from a repurposed oil well is approximately \$1.72 per square meter. This cost is significantly lower compared to a traditional geothermal heating system, which averages \$4.33/m². In twenty case studies from various countries, the energy efficiencies or coefficients of performance (COPs) for repurposing of decommissioned oil and gas wells into GWs ranged from 3.73 to 9.609, with an average of 8.51. Similarly, the thermal efficiency varied from 3.3 to 13.1 %, with an average of 9.4 % [31].

In a field study conducted in Chinese oil reservoirs, several critical aspects of power generation using co-produced hot oil and water from the LB reservoir (part of the Huabei oilfield owned by Petro-China) were examined. The average temperature of the reservoir formation and the

produced liquids was around 120 °C and 110 °C, respectively. One of the primary distinctions between geothermal and oil wells is their production rates, with oil wells typically yielding much less than GWs. The system was put into operation in April 2011 and had been running for several months. By the end of 2011, it had generated a cumulative energy output of approximately 31×10^4 kWh [77]. Sichuan Basin China has geothermal potential with seven major reservoirs. AOWs in this basin have a formation temperature ranging from 60 to 175 °C. The total resources of seven sets of thermal reservoirs have reached 4.26×10^{13} GJ, which is equivalent to $14,504 \times 10^8$ tons of standard coal [115]. The initial commercial project in North Dakota utilizes water at 98 °C with a flow rate of 30,000 bbl/day, resulting in a net power output of around 250 kW [116]. Moussa et al. [117] created a linear temperature gradient distribution map for 65,535 completed suspended oil and gas wells in the Western Canadian Sedimentary Basin. Using statistical methods, they estimated the BHT of these AOGWs. For a BHT of 85 °C and a production temperature of 50 °C at a depth of 3000 m, which is one kilometer away from residential areas, it is ultimately possible to provide 50 % of the GE for heating 2700 households, delivering 130 kW of geothermal power, which was estimated to increase to 59 kW after 25 years. Alimonti and Gnoni (2015) presented a case study on extending the lifespan of an oil field in northern Italy through the installation of a geothermal power plant. From an economic perspective, implementing the refurbishment project over the active field's lifetime can ensure the recovery of the investment cost for generating 400–500 kW of electrical power. The economic analysis indicates a payback period of 3.7 years [75].

There are three projects worldwide utilizing Organic Rankine Cycle (ORC) technology to generate electricity from high-temperature produced water in oil fields, as illustrated in Table 12 [39]. Two oil fields, OFA and OFB, in Colombia have established a shared geothermal energy production system to reduce carbon dioxide emissions and contribute to environmental preservation. This system uses ORC engines to convert the heat from fluids produced in the oil fields into electrical energy. According to thermodynamic calculations, the maximum theoretical potential for geothermal energy production is 2260 kW for the OFA field and 657 kW for the OFB field. The shared production system has resulted in a 19 % reduction in the carbon footprint for OFA and an 11 % reduction for OFB [118]. The summary of energy production cost from hydrocarbon fields for geothermal projects is presented in Table 13, in terms of expenses and project pricing at the field level [13]. For the generation of electricity from AOGWs using an OLS with a capacity of 3721.3 MW over 25 years, it has been evaluated that this capacity has a Net Present Value (NPV) of \$1,431,263,840 in U.S. dollars. The capital cost is \$12 million, operational costs are \$2 million, and the geothermal power cost is \$0.05/kWh, all calculated in U.S. dollars. These findings suggest that such projects are economically feasible [105].

The energy production parameters of power plants from several oil

Table 12
Summary of both direct and indirect utilization of oil wells [39].

Energy Extraction	Area	Heat/Power
Direct Utilizations	Shengli Oilfield, China	Heat: 10.3 GJ
Direct Utilizations	Huabei Oilfield, China	Heat: 7099 × 10 ⁹ GJ
Direct Utilizations	Daqing, Liaohe and Zhongyuan Oil fields, China	Heat: 2905 × 10 ⁹ GJ
Indirect Utilization	Wyoming Oilfield, USA	Power: 180 kW
Indirect Utilization	North Dakota, USA	Power: 250 kW
Indirect Utilization	Huabei Oilfield, China	Power: 310 kW

² Abandoned Oil Wells

Table 13

Assessment of the typical price range for GE in hydrocarbon fields worldwide [13].

Country	Hydrocarbon Site	Cost of energy US cents/kWh
China	–	9.3
Qatar	–	<5.3
Italy	Villafortuna Trecate	18.0
US	Banks field	10.95–19.15
US	Long Beach	11.0
US	Gulf Coast	8.0–11.0

reservoir examples and their conditions are presented in Table 14 [119]. Accurately estimating the energy potential when transitioning from AOGWs to GWs is essential. This process can be summarized in four key stages: energy potential assessment, economic analysis, environmental evaluation, and technological development. Accurately estimating energy production requires careful consideration of well logging, reservoir modeling, and well construction. Once these stages are completed, the final selection of a system and heat exchanger should prioritize compatibility with the well's specific conditions and cost-effectiveness. Additionally, environmental considerations must play a central role in the decision-making process. For instance, the potential dual-purpose use of carbon dioxide, such as for heat transfer or storage, offers an opportunity to enhance sustainability and reduce emissions. Given the growing urgency to address global energy demands and mitigate climate change, the strategic consideration of abandoned wells as a viable and forward-looking energy source is critical from both economic and political perspectives.

A crucial enabler of this transition is the adoption of forward-looking strategies by upstream oil and gas companies, particularly through the implementation of intelligent well-drilling technologies. These technologies can facilitate the seamless conversion of wells once they are abandoned, enabling their repurposing for renewable energy production. By integrating these innovative approaches, the energy landscape can shift away from fossil fuels and toward renewable energy solutions. This transition not only reduces pollution and minimizes environmental impact but also ensures the provision of sustainable energy for future generations. Ultimately, this approach addresses the growing global energy demand in a more balanced and environmentally responsible manner.

Repurposing AOGWs for GE extraction is economically viable, with costs significantly lower than new geothermal wells (e.g., \$1.72/m² vs. \$4.33/m² for heating [107]) and substantial energy outputs (e.g., 31 × 10⁴ kWh in Huabei oilfield [77]). Case studies, such as the Villafortuna-Trecate project (400–500 kW, 3.7-year payback [75]) and large-scale OLS projects (NPV \$1.43 billion [101]), demonstrate feasibility across diverse regions. Leveraging existing infrastructure reduces capital and decommissioning costs, while ORC technology enhances power generation. Future research should focus on scaling projects and integrating intelligent well technologies to optimize energy production and economic returns.

Table 14

Summary of energy production parameters of power plants from oil reservoirs [119].

	$T_2 - T_1$ (°F) ¹	T_i ² (°F)	Total Production (bbl day ⁻¹)	Power Plant Efficiency (%)	Power Plant Size (kW)	Geothermal Gradient (°F/ft)	Depth (ft)	NPV
Wilmington Oilfield	30	40	97,435	4.46	1015	0.034	4710	\$895,818
Inglewood Oilfield	19	40	19,727	4.45	205	0.035	3514	\$1,747,426
Long Beach Oilfield	31	40	124,683	4.42	1273	0.031	5273	\$8,262,660
Santa Fe Springs Oilfield	20	40	53,816	4.45	558	0.030	6744	\$3,143,029

¹ Temperature difference between reservoir and wellhead.

² Injection temperature

9. The way forward for future research and development of geothermal energy extraction from abandoned wells

This study focuses on the conversion of AOGWs into GWs by addressing technical, environmental, and economic challenges associated with repurposing existing well infrastructure for renewable energy extraction. By exploring innovative methods to enhance efficiency, reduce cost, and promote the sustainable use of these wells, this paper provides further practical insights into advancing GE as a viable and sustainable energy source. The following recommendations outline key areas for future research and development:

- 1. Integration of artificial intelligence (AI) and machine learning (ML) techniques.** The application of AI and ML in the conversion of AOGWs to GWs offers significant potential to enhance predictive modeling, optimize decision-making processes, and improve the overall operational efficiency of geothermal systems. Future research should focus on developing advanced algorithms for real-time data analysis, predictive maintenance, and resource optimization, enabling more accurate assessments of geothermal potential and operational performance.
- 2. Simulation of energy production from converted wells.** Conducting detailed simulations of energy production from AOGWs under both closed-loop and open-loop geothermal systems is essential for understanding the performance and energy yield of different system configurations. These simulations should incorporate site-specific geological and thermal characteristics to provide deeper insights into the applicability, efficiency, and scalability of various designs, ultimately guiding the selection of optimal configurations for diverse geological settings.
- 3. Development of hybrid systems for co-production of hydrocarbons and GE.** Investigating hybrid systems that enable the simultaneous extraction of residual hydrocarbons and GE could maximize the utility of AOGWs. Such systems would not only increase the overall energy output but also provide a transitional pathway for the oil and gas industry to adopt renewable energy technologies. Research should focus on optimizing system designs, operational strategies, and economic models to ensure the feasibility and sustainability of hybrid energy production.
- 4. Advancement of materials for wellbore integrity and heat transfer efficiency.** The development of advanced materials is critical for enhancing the structural integrity of wellbores and improving heat transfer efficiency in geothermal systems. Future studies should explore high-performance materials, such as advanced composites, thermal coatings, and corrosion-resistant alloys, to extend the lifespan of converted wells and optimize their thermal performance. Additionally, research into material behavior under high-temperature and high-pressure conditions is necessary to ensure long-term reliability.
- 5. Comprehensive environmental impact assessments.** Assessing the environmental impacts and sustainability of converting AOGWs into GWs is crucial for ensuring compliance with environmental standards and minimizing ecological risks. Future research should

focus on conducting detailed studies of potential effects on ecosystems, groundwater quality, and soil stability. These assessments should also evaluate the long-term environmental benefits of GE production, such as reductions in greenhouse gas emissions and land use compared to conventional energy sources.

6. **Integration of carbon capture and storage (CCS) with GE production.** Exploring the potential for combining GE production with carbon capture and storage (CCS) technologies could create dual-purpose systems that simultaneously produce renewable energy and mitigate CO₂ emissions. Research should focus on developing integrated systems that optimize the efficiency of both processes, assess their economic viability, and evaluate their potential to contribute to global decarbonization efforts.
7. **Economic feasibility studies in diverse geological settings.** Conducting economic feasibility studies specific to varying geological conditions is essential for identifying the most promising regions for cost-effective GE production. Future research should develop robust economic models that account for factors such as well depth, geothermal gradient, and infrastructure requirements. These models should also evaluate the cost-benefit trade-offs of repurposing existing wells versus drilling new GWs.
8. **Development of smart monitoring and control systems.** The implementation of smart monitoring and control systems using Internet of Things (IoT) technologies can significantly enhance the efficiency, safety, and automation of GW operations. Future research should focus on designing advanced sensor networks and real-time data analysis platforms to enable precise monitoring of well performance, early detection of operational issues, and adaptive control of geothermal systems. These technologies could also facilitate remote operation and maintenance, reducing operational cost and improving system reliability.

10. Conclusion

The conversion of abandoned oil and gas wells (AOGWs) into geothermal wells (GWs) offers a transformative opportunity for the oil and gas industry, characterized by cost-effectiveness, environmental sustainability, and the potential for enhanced energy generation. Successful transformation requires thorough planning, supportive governmental frameworks, and initial feasibility studies to assess each well's economic viability and energy production potential.

This research outlines effective energy extraction methods based on existing studies and cost analyses. Early planning in the well's lifecycle, from design to abandonment, is crucial to minimizing conversion costs and ensuring project success. Such proactive measures could pave the way for "smart" wells capable of simultaneously producing hydrocarbons and geothermal energy, representing a novel direction for the industry. Key considerations for maximizing this strategy include:

1. Repurposing AOGWs for geothermal energy extraction could provide a sustainable pathway to leverage existing oil and gas infrastructure, reducing capital costs and environmental impacts while meeting global energy demands through renewable energy integration.
2. Adaptations to above-ground facilities, improved low-temperature equipment, optimized well depth, and exploration of advanced materials significantly enhance geothermal energy recovery. Research into open-loop systems (OLS) and advanced numerical modeling is critical for economic viability.
3. Repurposing AOGWs has the potential to generate up to 31,000 kWh annually from single wells, lowering initial investments compared to developing new GWs.
4. The average cost of geothermal heating from repurposed wells is around \$1.72 per square meter, much lower than the \$4.33 for traditional systems, enhancing financial attractiveness for oil companies.
5. Geothermal energy production efficiency depends on geological parameters like geothermal gradient (approximately 30 °F per 1000 ft) and reservoir temperatures (60 °C to 175 °C). Effective management of fluid injection rates, heat transfer efficiency, and reservoir pressure is vital for maximizing energy output.
6. The choice of working fluids is essential for efficient heat transfer, requiring fluids with high thermal capacity, low viscosity, and stability under high-pressure conditions. Fluids like high-temperature water and light hydrocarbons are favored for their thermal properties.
7. Various systems, including open-loop and closed-loop (CLS) systems, each present unique challenges and efficiencies. Innovations enabling simultaneous geothermal and hydrocarbon production could enhance well utility.
8. The average investment for converting AOGWs to GWs is about \$12 million, with energy production costs around \$0.05 per kWh. This cost-effectiveness yields a favorable net present value (NPV) of approximately \$1.4 billion for large-scale projects.
9. Utilizing AOGWs for geothermal energy reduces greenhouse gas emissions and environmental pollution, supporting sustainable energy development while minimizing the ecological footprint of traditional well abandonment.

CRediT authorship contribution statement

Hamed Ojaghi: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Matin Shahin:** Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Mohammad Simjoo:** Writing – review & editing, Validation, Supervision, Project administration, Methodology, Formal analysis. **Mohammad Chahardowli:** Writing – review & editing, Validation, Supervision, Project administration, Methodology, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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